8. The velocity of an electron moving around a proton in an orbit of radius $5{\cdot}3 \times 10^{-11}$ m is $2{\cdot}2 \times 10^{6}$ ms^{-1}. Its frequency of revolution is : *1*

(A) $6{\cdot}6 \times 10^{15}$ Hz

(B) $5{\cdot}2 \times 10^{13}$ Hz

(C) $4{\cdot}3 \times 10^{12}$ Hz

(D) $3{\cdot}9 \times 10^{10}$ Hz

CBSE 26C

8. The velocity of an electron moving around a proton in an orbit of radius $5{\cdot}3 \times 10^{-11}$ m is $2{\cdot}2 \times 10^{6}$ ms^{-1}. Its frequency of revolution is : *1*

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(C) $4{\cdot}3 \times 10^{12}$ Hz

(D) $3{\cdot}9 \times 10^{10}$ Hz

CBSE 26C

(A) 6.6×10^{15} *Hz*

9. In alpha-particle scattering experiment, the impact parameter is minimum when the scattering angle is close to : *1*

(A) $45°$ (B) $90°$

(C) $135°$ (D) $180°$

CBSE 26C

9. In alpha-particle scattering experiment, the impact parameter is minimum when the scattering angle is close to : *1*

(A) 45° (B) 90°

(C) 135° (D) 180° CBSE 26C

(D) 180°

10. In Bohr's model of hydrogen atom, the angular momentum of electron in second orbit is equal to : *1*

(A) $\frac{h}{2\pi}$ (B) $\frac{h}{\pi}$

(C) $\frac{2h}{\pi}$ (D) $\frac{h}{4\pi}$

CBSE 26C

10. In Bohr's model of hydrogen atom, the angular momentum of electron in second orbit is equal to : *1*

(A) $\frac{h}{2\pi}$ (B) $\frac{h}{\pi}$

(C) $\frac{2h}{\pi}$ (D) $\frac{h}{4\pi}$

CBSE 26C

(B) $\frac{h}{\pi}$

8. The energy of an electron in an orbit in a hydrogen atom (Bohr model) is – 3·4 eV. Its angular momentum in this orbit is :

(A) $\frac{h}{\pi}$ (B) $\frac{h}{2\pi}$

(C) $\frac{h}{4\pi}$ (D) $\frac{2h}{\pi}$

CBSE 26C

8. The energy of an electron in an orbit in a hydrogen atom (Bohr model) is – 3·4 eV. Its angular momentum in this orbit is :

(A) $\frac{h}{\pi}$ (B) $\frac{h}{2\pi}$

(C) $\frac{h}{4\pi}$ (D) $\frac{2h}{\pi}$

CBSE 26C

(A) $\frac{h}{\pi}$

14. *Assertion (A) :* The ratio of the speeds of the electron in the second to that in the third orbit in Bohr's hydrogen atom is $\frac{3}{2}$.

Reason (R) : The speed of the electron in n^{th} permitted orbit in hydrogen atom varies proportional to $\frac{1}{n}$ i.e. $v_n \propto \frac{1}{n}$.

CBSE 26C

14. *Assertion (A) :* The ratio of the speeds of the electron in the second to that in the third orbit in Bohr's hydrogen atom is $\frac{3}{2}$.

Reason (R) : The speed of the electron in n^{th} permitted orbit in hydrogen atom varies proportional to $\frac{1}{n}$ i.e. $v_n \propto \frac{1}{n}$.

CBSE 26C

(A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

8. The Balmer series of hydrogen atom lies in :

(A) UV region (B) Infrared region

(C) Visible region (D) Microwave region

CBSE 26C

8. The Balmer series of hydrogen atom lies in :

(A) UV region (B) Infrared region

(C) Visible region (D) Microwave region

CBSE 26C

(C) Visible region

20. Calculate the distance of closest approach when an alpha particle of 4·50 MeV approaches in head-on position, a nucleus (Z = 79), stops and reverses its direction. CBSE 26C *2*

20. Calculate the distance of closest approach when an alpha particle of 4·50 MeV approaches in head-on position, a nucleus (Z = 79), stops and reverses its direction. CBSE 26C 2

$r = \frac{1}{4\pi\varepsilon_0} \times \frac{2Ze^2}{K}$	½
$K = 4.50\ MeV\ , Z = 79$	
$r = \frac{9\times10^9\times2\times79\times(1.6\,x\,10^{-19})^2}{4.50\times10^6\times\ 1.6\,x\,10^{-19}}$	1
$= 50.5 \times 10^{-15}$ m	
$= 50.5\ fm$	½

1. The radius of the innermost electron orbit of a hydrogen atom is 0·53 Å. The radius of orbit in the second excited state is :

(A) 3·5 Å (B) 4·77 Å

(C) 0·747 Å (D) 0·43 Å

CBSE 26C

1. The radius of the innermost electron orbit of a hydrogen atom is 0·53 Å. The radius of orbit in the second excited state is :

(A) 3·5 Å (B) 4·77 Å

(C) 0·747 Å (D) 0·43 Å

CBSE 26C

(B) 4.77 A^0

19. Find the ratio of energies of photons emitted during the transition of an electron in hydrogen atom from its : *2*

(i) second permitted energy level to the first permitted energy level, and

(ii) the highest permitted energy level to the first permitted energy level.

CBSE 26C

19. Find the ratio of energies of photons emitted during the transition of an electron in hydrogen atom from its : *2*

(i) second permitted energy level to the first permitted energy level, and

(ii) the highest permitted energy level to the first permitted energy level.

CBSE 26C

(i) Use $E_n = -\frac{13.6}{n^2}$	½
$\Delta E_{21} = E_2 - E_1$	
$= -13.6\left(\frac{1}{4} - 1\right)$	
$\Delta E_{21} = 13.6 \times \frac{3}{4}$ eV ...(1)	½
$\Delta E_{\infty 1} = -13.6\left(\frac{1}{\infty} - 1\right)$	½
$\Delta E_{\infty 1} = 13.6$ eV ...(2)	
From (1) and (2),	
$\frac{\Delta E_{21}}{\Delta E_{\infty 1}} = \frac{3}{4}$	½

10. The energy of an electron in an orbit in hydrogen atom is – 3·4 eV. Its angular momentum in the orbit will be :

(A) $\frac{3h}{2\pi}$ (B) $\frac{2h}{\pi}$

(C) $\frac{h}{\pi}$ (D) $\frac{h}{2\pi}$

CBSE 26

10. The energy of an electron in an orbit in hydrogen atom is – 3·4 eV. Its angular momentum in the orbit will be :

(A) $\frac{3h}{2\pi}$

(B) $\frac{2h}{\pi}$

(C) $\frac{h}{\pi}$

(D) $\frac{h}{2\pi}$

CBSE 26

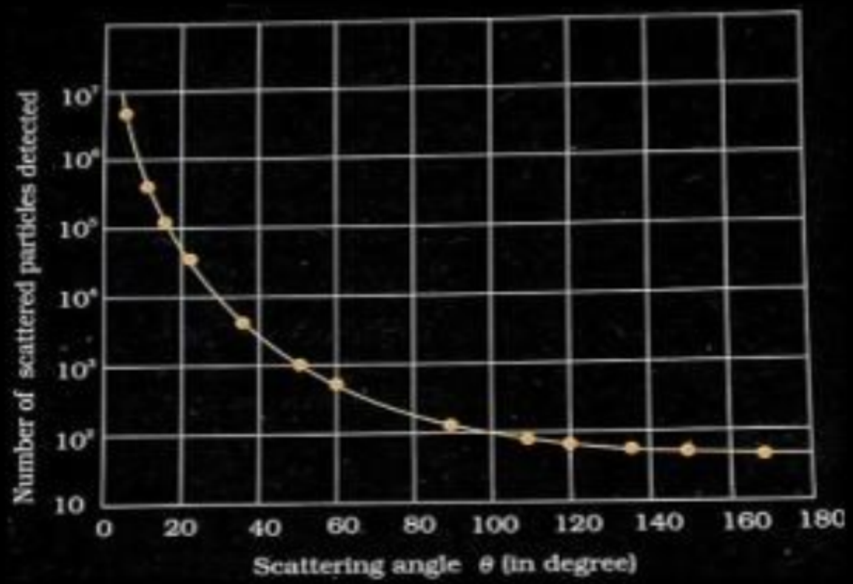
(C) $\frac{h}{\pi}$

(b) (i) Draw the number of scattered particles versus the scattering angle graph for scattering of alpha particles by a thin foil. Write two important conclusions that can be drawn from this plot.

(ii) If Bohr's quantization postulate (angular momentum = $\frac{nh}{2\pi}$) is a basic law of nature, it should be equally valid for the case of planetary motion also. Why, then, do we never speak of quantization of orbits of planets around the Sun ? Explain. *3*

CBSE 26

(i)



1

Important conclusions:

1.Most of the space in the atom is empty.

2. Most part of mass of atom and its positive charge were tightly concentrated at its centre.

1

Note: No marks to be deducted for scale on the axis

(ii) According to Bohr Quantisation postulate

$L=mvr = nh/2\pi$

Since mass of planets is very large as compared to mass of atomic particles, angular momentum (L) will be very large.

1

Therefore, Principal quantum number (n) will be very large.

For large quantum number. the difference in successive energy levels will be negligible. Hence quantization fails.

12. The energy of an electron revolving in the n^{th} orbit in Bohr model of hydrogen atom is proportional to :

(A) n^2 (B) n

(C) $\frac{1}{n}$ (D) $\frac{1}{n^2}$

CBSE 26

12. The energy of an electron revolving in the n^{th} orbit in Bohr model of hydrogen atom is proportional to :

(A) n^2 (B) n

(C) $\frac{1}{n}$ (D) $\frac{1}{n^2}$

CBSE 26

(D) $\frac{1}{n^2}$

11. Paschen series in spectrum of hydrogen atom lies in :

(A) infrared region

(B) ultraviolet region

(C) visible region

(D) partly in ultraviolet region and partly in visible region

CBSE 26

11. Paschen series in spectrum of hydrogen atom lies in :

(A) infrared region

(B) ultraviolet region

(C) visible region

(D) partly in ultraviolet region and partly in visible region

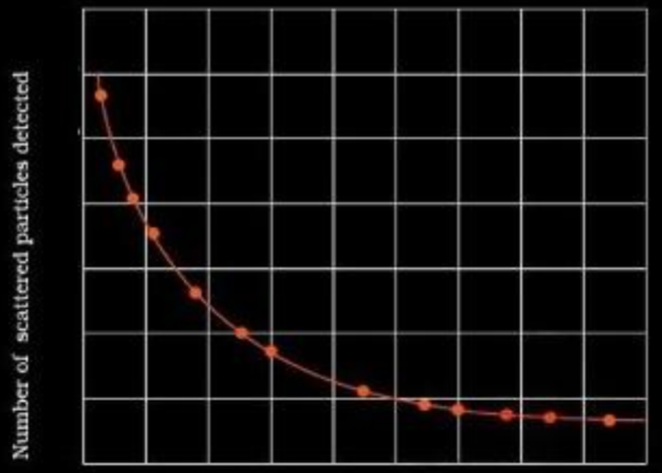
CBSE 26

(A) Infrared region

20. Draw the graph showing the variation of number of scattered alpha particles (N) detected with scattering angle (θ) in Geiger-Marsden experiment. Infer two conclusions from the graph. CBSE 26 *2*

20. Draw the graph showing the variation of number of scattered alpha particles (N) detected with scattering angle (θ) in Geiger-Marsden experiment. Infer two conclusions from the graph. *2*

CBSE 26



1

Conclusions from the graph

1. Many of the α-particles pass through the foil, without suffering any collisions.
2. Only a few incident α-particles scatter by more than 1°.
3. About 1 in 8000 α-particles deflect by more than 90°.

½ + ½

(Any Two)

24. In Bohr model of hydrogen atom, calculate : *3*

(a) the radius of the innermost electron orbit, and

(b) the speed of electron in this orbit.

CBSE 26

24. In Bohr model of hydrogen atom, calculate : *3*

(a) the radius of the innermost electron orbit, and

(b) the speed of electron in this orbit.

CBSE 26

(a) The radius of innermost electron orbit is given by, ½

$$r = -\frac{e^2}{8\pi\varepsilon_0 E} \quad \text{------(1)}$$

where E = Energy of electron in hydrogen atom in innermost orbit (n = 1).

Using formula, $E_n = -\frac{13.6}{n^2}\text{eV}$

For n =1, $E_1 = -\frac{13.6}{1^2}\text{eV} = -13.6 \times 1.6 \times 10^{-19}\text{J} = -2.2\times10^{-18}\text{J}$ ------(2)

Substituting the value of E for n =1 from equation (2) into (1), we get

$$r = -\frac{9\times10^9 \times (1.6\times10^{-19})^2}{2(-2.2\times10^{-18})}$$ ½

$$= 5.3\times10^{-11}\text{m}$$ ½

(b) The speed of electron in this orbit is

$$v = \frac{e}{\sqrt{4\pi\varepsilon_0 mr}}$$ ½

$$= \frac{1.6\times10^{-19}}{\sqrt{\frac{1}{9\times10^9}\times9.1\times10^{-31}\times5.3\times10^{-11}}}$$ ½

$$= 2.2\times10^6\text{m/s}$$ ½

Alternatively

The radius of innermost electron orbit in H – atom is given by

$$r_n = \frac{n^2}{Z}r_0$$

where $r_0 = 0.529\text{A}^0$

Using n =1, Z=1,

$r_1 = 0.529\times10^{-10}\text{m}$

Speed of electron in this orbit,

$$v_n = \frac{1}{137}\frac{c}{n} \text{ where } c = 3\times10^8\text{m/sec}$$

$$= \frac{1}{137}\frac{3\times10^8}{1} = 2.19\times10^6\text{m/s}$$

10. The ratio of the de Broglie wavelengths associated with the electron revolving in the first and third orbits in hydrogen atom is :

(A) 2

(B) $\frac{1}{2}$

(C) 3

(D) $\frac{1}{3}$

CBSE 26

10. The ratio of the de Broglie wavelengths associated with the electron revolving in the first and third orbits in hydrogen atom is :

(A) 2

(B) $\frac{1}{2}$

(C) 3

(D) $\frac{1}{3}$

CBSE 26

(D) $\frac{1}{3}$

28. A deuterium nucleus and an alpha particle approach a target nucleus separately in head-on position. Find the ratio of their distance of closest approach to the nucleus, when both have :

(a) the same velocity, and

(b) the same momentum.

CBSE 26

3

28. A deuterium nucleus and an alpha particle approach a target nucleus separately in head-on position. Find the ratio of their distance of closest approach to the nucleus, when both have :

(a) the same velocity, and

(b) the same momentum.

CBSE 26 *3*

(a) $\frac{1}{2}mv^2 = \frac{kq_1q_2}{d}$

$d \propto \frac{q}{m}$ (for same velocity) ½

$\frac{d_{deutron}}{d_\alpha} = \frac{q_{deutron}}{q_\alpha} \times \frac{m_\alpha}{m_{deutron}}$

$= \frac{e}{2e} \times \frac{4m}{2m}$ ½

$= 1$ ½

$= d_{deutron} : d_\alpha = 1:1$

(b) $\frac{p^2}{2m} = \frac{kq_1q_2}{d}$ ½

$d \propto q\,m$ (for same momemtu

$\frac{d_{deutron}}{d_\alpha} = \frac{q_{deutron} \times m_{deutron}}{q_\alpha \times m_\alpha}$

$= \frac{e \times 2m}{2e \times 4m}$ ½

$= \frac{1}{4}$ ½

$d_{deutron} : d_\alpha = 1 : 4$

7. Suppose there is an atom that contains five energy levels. How many transitions are possible in which photons are emitted ? **1**

(A) 4 (B) 5

(C) 10 (D) 15

CBSE 26

7. Suppose there is an atom that contains five energy levels. How many transitions are possible in which photons are emitted ? **1**

(A) 4 (B) 5

(C) 10 (D) 15

CBSE 26

(C) 10

10. In Bohr's model of hydrogen atom, the potential energy of the electron in the ground state is – **1**

(A) –27.2 eV (B) 27.2 eV

(C) –13.6 eV (D) 13.6 eV

CBSE 26

10. In Bohr's model of hydrogen atom, the potential energy of the electron in the ground state is – **1**

(A) –27.2 eV (B) 27.2 eV

(C) –13.6 eV (D) 13.6 eV

CBSE 26

(A) -27.2eV

27. The total energy of an electron is an orbit in Bohr model of hydrogen atom is –3.4 eV. Calculate :

(i) the angular momentum of the electron, in this orbit and

(ii) the potential energy of the electron, in this orbit. **3**

CBSE 26

27. The total energy of an electron is an orbit in Bohr model of hydrogen atom is –3.4 eV. Calculate :

(i) the angular momentum of the electron, in this orbit and

(ii) the potential energy of the electron, in this orbit. **3**

CBSE 26

(a) Total energy, $E_n = \frac{-13.6}{n^2}$ eV ½

$$\Rightarrow \quad n^2 = \frac{-13.6}{-3.4} = 4$$

$$n = 2$$

Angular momentum, $L = \frac{nh}{2\pi}$ ½

$$= \frac{(2)(6.63\times10^{-34})}{(2)(3.14)}$$

$$= 2.11\times10^{-34}\text{Js}$$ ½

(b) Potential energy = 2×Total Energy ½

$$= 2\times(-3.14)$$ ½

$$= -6.8\text{eV}$$ ½

4. In Bohr model of hydrogen atom, for large values of n, the distance between the consecutive orbits is proportional to **1**

(A) $\sqrt{n}$ (B) n

(C) n^2 (D) n^3

CBSE 26

4. In Bohr model of hydrogen atom, for large values of n, the distance between the consecutive orbits is proportional to **1**

(A) $\sqrt{n}$ (B) n

(C) n^2 (D) n^3

CBSE 26

(B) n

12. In Bohr model of hydrogen atom, the electron makes a transition from n = 5 to n = 1 state. As a result, a photon of wavelength λ is emitted in the process. The wavelength of the photon emitted when an electron makes a transition from energy level n = 5 to n = 2 will be **1**

(A) $\frac{8}{7}\lambda$

(B) $\frac{24}{7}\lambda$

(C) $\frac{16}{7}\lambda$

(D) $\frac{32}{7}\lambda$

CBSE 26

12. In Bohr model of hydrogen atom, the electron makes a transition from n = 5 to n = 1 state. As a result, a photon of wavelength λ is emitted in the process. The wavelength of the photon emitted when an electron makes a transition from energy level n = 5 to n = 2 will be **1**

(A) $\frac{8}{7}\lambda$ (B) $\frac{24}{7}\lambda$

(C) $\frac{16}{7}\lambda$ (D) $\frac{32}{7}\lambda$

CBSE 26

(D) $\frac{32}{7}\lambda$

26. (a) State Bohr's second postulate and mention its significance. **3**

(b) Prove that, in Bohr model of hydrogen atom as principal quantum number n becomes large, the energy levels get closer and closer. CBSE 26

26. (a) State Bohr's second postulate and mention its significance. **3**

(b) Prove that, in Bohr model of hydrogen atom as principal quantum number n becomes large, the energy levels get closer and closer. CBSE 26

(a)	The electron revolves around the nucleus only in those orbits for which angular momentum is some integral multiple of $\frac{h}{2\pi}$ where h is Planck's constant.	1
	Alternatively:	
	$L = n\left(\frac{h}{2\pi}\right)$	
	• It defines stable / stationary orbits	$^1/_2$
	Alternatively	
	Angular momentum is quantised	
(b)	$E_n = \frac{-13.6}{n^2}$ eV	$^1/_2$
	For any two consecutive energy levels, the energy difference is: $\Delta E = E_n - E_{n-1}$	
	$\Delta E = 13.6\left[\frac{1}{(n-1)^2} - \frac{1}{n^2}\right]$	$^1/_2$
	$\Delta E = 13.6\ \frac{(2n-1)}{n^2(n-1)^2}$	
	For large value of n	
	$\Delta E = \frac{27.2}{n^3}$ eV	
	Thus as n increase, ΔE decrease i.e., energy levels get closer and closer.	$^1/_2$

3. In Bohr model of hydrogen atom, the value of potential energy of an electron in n^{th} orbit varies with 'n' as **1**

(A) $\frac{1}{n^2}$ (B) $\frac{1}{n}$

(C) n (D) n^2

CBSE 26

3. In Bohr model of hydrogen atom, the value of potential energy of an electron in n^{th} orbit varies with 'n' as **1**

(A) $\frac{1}{n^2}$ (B) $\frac{1}{n}$

(C) n (D) n^2

CBSE 26

(A) $\frac{1}{n^2}$

5. In Bohr model of hydrogen atom, an electron makes a transition from n = 4 state to n = 1 state and a photon of frequency v is emitted. The frequency of photon emitted when an electron makes a transition from n = 4 state to n = 2 state in the same model is **1**

(A) $\frac{v}{3}$

(B) $\frac{v}{4}$

(C) $\frac{v}{5}$

(D) $\frac{v}{6}$

CBSE 26

5. In Bohr model of hydrogen atom, an electron makes a transition from n = 4 state to n = 1 state and a photon of frequency ν is emitted. The frequency of photon emitted when an electron makes a transition from n = 4 state to n = 2 state in the same model is **1**

(A) $\frac{\nu}{3}$ (B) $\frac{\nu}{4}$

(C) $\frac{\nu}{5}$ (D) $\frac{\nu}{6}$

CBSE 26

(C) $\frac{\nu}{5}$

23. Calculate the de Broglie wavelength associated with an electron revolving in the second excited state of hydrogen atom. The ground state energy of the hydrogen atom is – 13.6 eV. Take $m_e = 9 \times 10^{-31}$ kg, $h = 6.6 \times 10^{-34}$ J.s. **3**

CBSE 26

23. Calculate the de Broglie wavelength associated with an electron revolving in the second excited state of hydrogen atom. The ground state energy of the hydrogen atom is – 13.6 eV. Take $m_e = 9 \times 10^{-31}$ kg, $h = 6.6 \times 10^{-34}$ J.s. **3**

CBSE 26

Total energy of electron in second exited state of hydrogen atom	
$E_n = \frac{-13.6}{n^2} eV$	½
$E_3 = \frac{-13.6}{3^2} eV$	
$E_3 = -1.51 eV$	½
$\therefore$ Kinetic energy of electron is this state, $K = 1.51 eV$	½
de Broglie wavelength	
$\lambda = \frac{h}{\sqrt{2mK}}$	½
$= \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9 \times 10^{-31} \times 1.51 \times 1.6 \times 10^{-19}}}$	½
$\lambda \approx 10^{-9} m = 1nm$	½

9. In Bohr model of hydrogen atom, an electron is moving in an orbit, with total energy –3.4 eV. Its angular momentum will be **1**

(A) 1.05×10^{-34} J s

(B) 6.63×10^{-34} J s

(C) 3.31×10^{-34} J s

(D) 2.1×10^{-34} J s

CBSE 26

9. In Bohr model of hydrogen atom, an electron is moving in an orbit, with total energy –3.4 eV. Its angular momentum will be **1**

(A) 1.05×10^{-34} J s

(B) 6.63×10^{-34} J s

(C) 3.31×10^{-34} J s

(D) 2.1×10^{-34} J s

CBSE 26

(D) 2.1×10^{-34} Js

10. According to Bohr model of hydrogen atom, the time period of an electron in its n^{th} orbit is proportional to **1**

(A) n (B) n^2

(C) n^3 (D) n^4

CBSE 26

10. According to Bohr model of hydrogen atom, the time period of an electron in its n^{th} orbit is proportional to **1**

(A) n (B) n^2

(C) n^3 (D) n^4

CBSE 26

(C) n^3

8. The 'distance of closest approach' of an alpha-particle is 'd' when it moves with a velocity v head-on towards the target nucleus. If the velocity of alpha particle is halved, the new 'distance of closest approach' will be – **1**

(A) $\frac{d}{2}$ (B) 2d

(C) $\frac{d}{4}$ (D) 4d

CBSE 26

8. The 'distance of closest approach' of an alpha-particle is 'd' when it moves with a velocity v head-on towards the target nucleus. If the velocity of alpha particle is halved, the new 'distance of closest approach' will be – **1**

(A) $\frac{d}{2}$ (B) 2d

(C) $\frac{d}{4}$ (D) 4d

CBSE 26

(D) 4d

16. **Reason (A) :** In Bohr model of hydrogen atom, the energy levels are discrete and quantised. **1**

Reason (R) : In a hydrogen atom, the electrostatic force on the electron provides the necessary centripetal force to it to revolve around the nucleus.

CBSE 26

16. **Reason (A) :** In Bohr model of hydrogen atom, the energy levels are discrete and quantised. **1**

Reason (R) : In a hydrogen atom, the electrostatic force on the electron provides the necessary centripetal force to it to revolve around the nucleus.

CBSE 26

16. **Reason (A) :** In Bohr model of hydrogen atom, the energy levels are discrete and quantised. **1**

Reason (R) : In a hydrogen atom, the electrostatic force on the electron provides the necessary centripetal force to it to revolve around the nucleus.

CBSE 26

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A)

6. The distance-of-closest-approach for an alpha particle is 'd' when it moves head-on with speed v towards a target nucleus. If the alpha particle is replaced by a proton moving with the same speed, the new distance-of-closest-approach will be 1

(A) 2d

(B) d

(C) $\frac{d}{2}$

(D) $\frac{d}{\sqrt{2}}$

CBSE 26

6. The distance-of-closest-approach for an alpha particle is 'd' when it moves head-on with speed v towards a target nucleus. If the alpha particle is replaced by a proton moving with the same speed, the new distance-of-closest-approach will be 1

(A) 2d

(B) d

(C) $\frac{d}{2}$

(D) $\frac{d}{\sqrt{2}}$

CBSE 26

(A) 2d

8. The ratio of the potential energy to the kinetic energy of an electron in n^{th} orbit of Bohr model of hydrogen atom is **1**

(A) $-\frac{1}{2}$ (B) $\frac{1}{2}$

(C) 2 (D) –2

CBSE 26

8. The ratio of the potential energy to the kinetic energy of an electron in n^{th} orbit of Bohr model of hydrogen atom is **1**

(A) $-\frac{1}{2}$ (B) $\frac{1}{2}$

(C) 2 (D) –2

CBSE 26

(D) - 2

2. In Bohr model of the hydrogen atom, the kinetic energy of an electron in $n = 2$ orbit is :

(A) 1·7 eV

(B) 3·4 eV

(C) 6·8 eV

(D) 10·2 eV

CBSE 2025 C

2. In Bohr model of the hydrogen atom, the kinetic energy of an electron in n = 2 orbit is :

(A) 1·7 eV (B) 3·4 eV

(C) 6·8 eV (D) 10·2 eV

CBSE 2025 C

(B) 3.4 eV

2. In Bohr model of the hydrogen atom, the potential energy of electron in the ground state is :

(A) 13·6 eV

(B) – 13·6 eV

(C) 27·2 eV

(D) – 27·2 eV

CBSE 2025 C

2. In Bohr model of the hydrogen atom, the potential energy of electron in the ground state is :

(A) 13·6 eV

(B) – 13·6 eV

(C) 27·2 eV

(D) – 27·2 eV

CBSE 2025 C

(D) –27.2 eV

2. In a Rutherford scattering experiment, when an alpha particle of mass m_1 approaches a target nucleus of charge Ze and mass m_2, the distance of the closest approach is d_0. The energy of the particle is :

(A) directly proportional to Z

(B) inversely proportional to Z

(C) directly proportional to mass m_2

(D) directly proportional to $m_1 m_2$

CBSE 2025 C

2. In a Rutherford scattering experiment, when an alpha particle of mass m_1 approaches a target nucleus of charge Ze and mass m_2, the distance of the closest approach is d_0. The energy of the particle is :

(A) directly proportional to Z

(B) inversely proportional to Z

(C) directly proportional to mass m_2

(D) directly proportional to m_1m_2

CBSE 2025 C

(A) directly proportional to Z.

24. Write the three postulates of Bohr model of hydrogen atom. Show that, in this model, the frequency of revolution of an electron in its n^{th} orbit is proportional to $\left(\frac{1}{n^3}\right)$.

CBSE 2025 C

3

Bohr's Postulates

(i) An electron in an atom could revolve in certain stable orbits without the emission of radiant energy. ½

(ii) Electron revolves around the nucleus only in those orbits for which the angular momentum is some integral multiple of $h/2\pi$. ½

(iii) An electron might make a transition from one of its specified non-radiating orbits to another of lower energy. When it does so, a photon is emitted having energy equal to the energy difference between the initial and final states. The frequency of the emitted photon is then given by ½

$$h\nu = E_i - E_f$$

As frequency of revolution of electron is given by

$$\nu = \frac{v}{2\pi r} \quad \text{.....(1)}$$

In Bohr's model the velocity of the electron in the nth orbit is

$$v_n = \frac{e^2}{2\varepsilon_0 nh}$$

$$v_n \ \alpha \ \frac{1}{n} \quad \text{.....(2)} \qquad ½$$

and radius of the electron in the nth orbit is

$$r_n = \frac{\varepsilon_0 n^2 h^2}{\pi m e^2}$$

$$r_n \ \alpha \ n^2 \quad \text{----------(3)} \qquad ½$$

Using (3) and (2) in (1)

$$\nu \ \alpha \ \frac{1}{n^3} \qquad ½$$

17. Suppose that an electron is revolving around the nucleus of a hydrogen atom in the orbit of radius 0·48 Å. Find the magnitude and direction of the electrostatic force acting on the electron. *2*

CBSE 2025 C

17. Suppose that an electron is revolving around the nucleus of a hydrogen atom in the orbit of radius 0·48 Å. Find the magnitude and direction of the electrostatic force acting on the electron. *2*

CBSE 2025 C

$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_1}{r^2}$	½
$F = 9 \times 10^9 \times \frac{1.6 \times 10^{-19} \times 1.6 \times 10^{-19}}{4.8 \times 4.8 \times 10^{-22}}$	½
$= 10^{-7}$ N	½
Towards nucleus	½

20. (a) An electron makes a transition from (i) state n = 3 to state n = 2, and (ii) state n = 2 to state n = 1, in hydrogen atom (Bohr model). Find the ratio of the wavelengths of radiations emitted in the two cases. CBSE 2025 C *2*

20. (a) An electron makes a transition from (i) state n = 3 to state n = 2, and (ii) state n = 2 to state n = 1, in hydrogen atom (Bohr model). Find the ratio of the wavelengths of radiations emitted in the two cases. CBSE 2025 C *2*

$\frac{1}{\lambda}=R\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]$	½
$\frac{1}{\lambda_1}=R\left[\frac{1}{(2)^2}-\frac{1}{(3)^2}\right]=\frac{5}{36}R$	½
$\frac{1}{\lambda_2}=R\left[\frac{1}{(1)^2}-\frac{1}{(2)^2}\right]=\frac{3}{4}R$	½
$\frac{\lambda_2}{\lambda_1}=\frac{5}{27}$	½

10. The minimum energy required to free the electron from the ground state of the hydrogen atom is :

(A) 3·4 eV

(B) less than 3·4 eV

(C) 13·6 eV

(D) between 13·6 eV and 3·4 eV

CBSE 2025

10. The minimum energy required to free the electron from the ground state of the hydrogen atom is :

(A) 3·4 eV

(B) less than 3·4 eV

(C) 13·6 eV

(D) between 13·6 eV and 3·4 eV

CBSE 2025

(C) 13.6 eV

19. Define the 'distance of closest approach'. An α-particle of kinetic energy K is bombarded on a thin gold foil. Derive an expression for the 'distance of closest approach'. CBSE 2025 *2*

19. Define the 'distance of closest approach'. An α-particle of kinetic energy K is bombarded on a thin gold foil. Derive an expression for the 'distance of closest approach'. *2*

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The distance between the α particle and the target nucleus when whole kinetic energy of an α particle gets converted into potential energy.	½
Alternatively: It is the distance from the nucleus at which alpha particle stops momentarily and then begins to retrace its path.	
At distance of closest approach $K = U$ $K = \frac{(2e)(Ze)}{4\pi\varepsilon_0 d}$	1
$\therefore d = \frac{2Ze^2}{4\pi\varepsilon_0 K}$	½

28. Write the mathematical forms of three postulates of Bohr's theory of the hydrogen atom. Using them prove that, for an electron revolving in the n^{th} orbit,

CBSE 2025

(a) the radius of the orbit is proportional to n^2, and

(b) the total energy of the atom is proportional to $\left(\frac{1}{n^2}\right)$. *3*

Mathematical form of postulates of Bohr's Theory

(i) $E_n = \frac{-13.6}{n^2} eV$ ½

Alternatively : Electron revolve in stable orbits with definite energy called stationary orbits.

(ii) $L = mvr = \frac{nh}{2\pi}$ ½

(iii) $h\nu = E_f - E_i$ ½

(a) $\frac{mv^2}{r} = \frac{Ze^2}{r^2}$ -----(1) ½

$mvr = \frac{nh}{2\pi}$ -----(2)

Solving (1) & (2)

$r = n^2 \left(\frac{h}{2\pi}\right)^2 \frac{4\pi\varepsilon_0}{me^2}$ ----(3) ½

Since energy in the orbit $E_n = \frac{-e^2}{8\pi\varepsilon_0 r}$

Using eq (3) $E_n = \frac{-me^4}{8n^2\varepsilon_0{}^2h^2}$ ½

or $E_n \propto \frac{1}{n^2}$

10. Out of the four options given, in which transition will the emitted photon have the maximum wavelength ?

(A) $n = 4$ to $n = 3$

(B) $n = 3$ to $n = 2$

(C) $n = 2$ to $n = 1$

(D) $n = 3$ to $n = 1$

CBSE 2025

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(A) $n = 4$ to $n = 3$ (B) $n = 3$ to $n = 2$

(C) $n = 2$ to $n = 1$ (D) $n = 3$ to $n = 1$

CBSE 2025

(A) n=4 to n=3

20. An alpha particle and a deuterium ion are accelerated through the same potential difference. These are then directed towards a target nucleus to make head-on collision. It is observed that their distance of closest approach is the same. Justify it theoretically. CBSE 2025 *2*

20. An alpha particle and a deuterium ion are accelerated through the same potential difference. These are then directed towards a target nucleus to make head-on collision. It is observed that their distance of closest approach is the same. Justify it theoretically. CBSE 2025 *2*

For a given nucleus, the distance of closest approach for a charged particle depends only on the accelerating potential difference. Since both α -particle and a deuterium ion are accelerated through same potential difference, therefore distance of closest approach will be same for both. **Note:** Award 1 mark if a student proves it mathematically.	2

27. In Bohr model of hydrogen atom, an electron is revolving in second orbit. Find the value of :

(i) angular momentum of electron,

(ii) radius of the orbit, and

(iii) kinetic energy of electron.

CBSE 2025

Take radius of first orbit of hydrogen atom as 0·5 Å. *3*

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CBSE 2025

Take radius of first orbit of hydrogen atom as 0·5 Å. *3*

(i)	$L=\frac{nh}{2\pi}$	½
	$for \ \ n=2$	
	$L=\frac{2\times6.63\times10^{-34}}{2\times3.14}$	
	$=2.11\times10^{-34}\ kg\,m^2\,s^{-1}$	½
(ii)	$r_n=n^2r_0$	½
	$r_2=4(0.5\,\mathring{A})$	
	$r_2=2\,\mathring{A}$	½
(iii)	Total energy of electron $=-\frac{13.6}{n^2}$ eV	½
	$E=-3.4\text{eV} \quad (n=2)$	
	$K=-E$	
	$K=3.4\text{eV}$	½

25. Describe briefly Geiger-Marsden scattering experiment. Depict the graph showing the variation of number of scattered particles detected with the scattering angle. How did this graph lead to the discovery of the nucleus ? *3*

CBSE 2025

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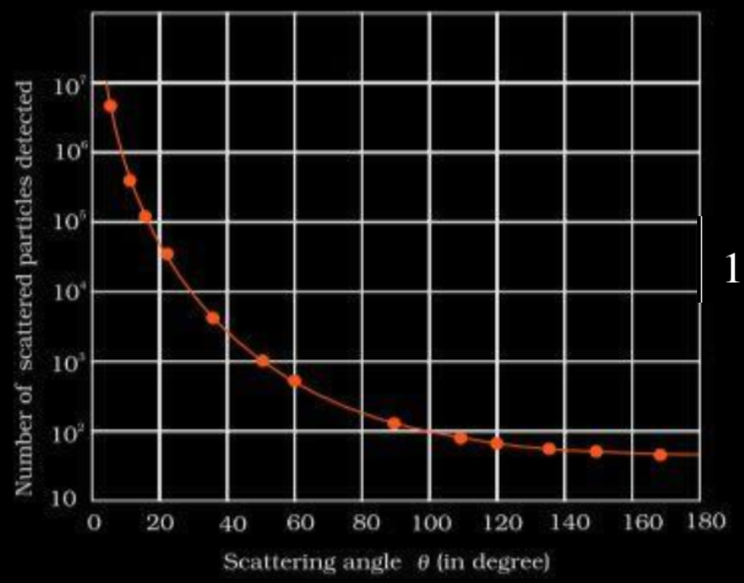
CBSE 2025



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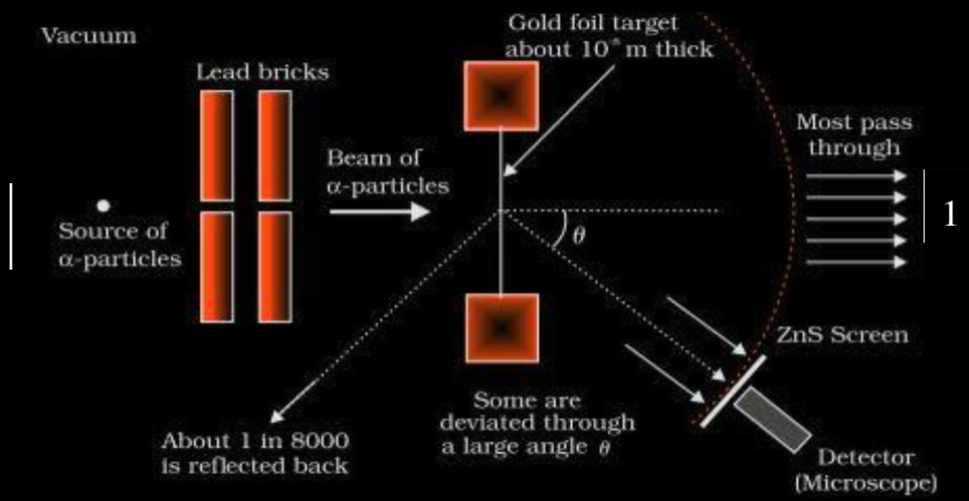

1

A small fraction of number of incident α -particles rebound back. It indicates that the number of α -particles undergoing head on collision is very small. From this we conclude that the positive charge is concentrated in a very small volume called nucleus.

1

27. How is the necessary force provided to an electron to keep it moving in a circular orbit according to Bohr model of hydrogen atom ? Derive an expression for the total energy of an electron moving in an orbit of radius r in hydrogen atom. Give the significance of negative sign in this expression.

CBSE 2025

3

The electrostatic force of attraction between the electrons and the nucleus provides the necessary centripetal force required to an electron to revolve in the orbit.	½
$\frac{mv^2}{r}=\frac{e^2}{4\pi\varepsilon_0 r^2}$ -------(1) (Z = 1 for hydrogen atom)	½
Kinetic energy of the electron $K=\frac{1}{2}mv^2$	
$K=\frac{e^2}{8\pi\varepsilon_0 r}$ (from eq (1))	½
Potential energy of the electron $U=\frac{-e^2}{4\pi\varepsilon_0 r}$ $\left(\because U=\frac{q_1 q_2}{4\pi\varepsilon_0 r}\right)$	½
Total energy of the electron E = K + U $E=\frac{-e^2}{8\pi\varepsilon_0 r}$	½
Note: Full credit of this part should be given if a student shows this derivation using alternative method Negative sign signifies that electron is bound to the nucleus OR force is attractive.	½

10. Which one out of the following transitions of an electron in Bohr's model of hydrogen atom will emit a photon with the greatest frequency ? **1**

(A) $n = 3$ to $n = 1$ (B) $n = 3$ to $n = 2$

(C) $n = 4$ to $n = 3$ (D) $n = 4$ to $n = 2$

CBSE 2025

10. Which one out of the following transitions of an electron in Bohr's model of hydrogen atom will emit a photon with the greatest frequency ? 1

(A) n = 3 to n = 1
(B) n = 3 to n = 2
(C) n = 4 to n = 3
(D) n = 4 to n = 2

CBSE 2025

(A) n=3 to n=1

20. The total energy of an electron in the hydrogen atom in the ground state is –13.6 eV. Find : **2**

(a) the potential energy of the electron.

(b) the kinetic energy of the electron.

(c) the ratio of the magnitude of the total energy to the kinetic energy of the electron.

CBSE 2025

20. The total energy of an electron in the hydrogen atom in the ground state is –13.6 eV. Find : **2**

(a) the potential energy of the electron.

(b) the kinetic energy of the electron.

(c) the ratio of the magnitude of the total energy to the kinetic energy of the electron.

CBSE 2025

(a) $U = 2E = 2\times(-13.6) = -27.2\,eV$	½
(b) $K = -E = -(-13.6)eV = +13.6\,eV$	½
(c) $\frac{E}{K} = \frac{13.6eV}{13.6eV} = 1$	1

27. State postulates of Bohr model of hydrogen atom. Using Bohr's second postulate, show that the circumference of n^{th} orbit in hydrogen atom is n times the de Broglie wavelength associated with the electron revolving in it. CBSE 2025 **3**

27. State postulates of Bohr model of hydrogen atom. Using Bohr's second postulate, show that the circumference of n^{th} orbit in hydrogen atom is n times the de Broglie wavelength associated with the electron revolving in it. CBSE 2025 **3**

Bohr's Postulates

(i) an electron in an atom could revolve in certain stable orbits without the emission of radiant energy. ½

(ii) The electron revolves around the nucleus only in those orbits for which the angular momentum is some integral multiple of $\frac{h}{2\pi}$. ½

Alternatively, $L = \frac{nh}{2\pi} = mvr$

(iii) An electron might make a transition from one of its specified non radiating orbits to another of lower energy , if it does so a photon is emitted having energy equal to the energy difference between the intial and final states.The frequency of emitted photon is then given by $E_i - E_f = h\upsilon$ ½

From second postulates

$$mv_n r_n = \frac{nh}{2\pi}$$ ½

$$2\pi r_n = \frac{nh}{mv_n}$$ ½

$$2\pi r_n = n\lambda$$ ½

14. *Assertion (A) :* In Rutherford's alpha particle scattering experiment, the presence of only few alpha particles at angle of scattering π led him to the discovery of nucleus.

Reason (R) : The size of nucleus is approximately 10^{-5} times the size of an atom and therefore only few alpha particles are rebounded.

CBSE 2025

14. *Assertion (A) :* In Rutherford's alpha particle scattering experiment, the presence of only few alpha particles at angle of scattering π led him to the discovery of nucleus.

Reason (R) : The size of nucleus is approximately 10^{-5} times the size of an atom and therefore only few alpha particles are rebounded.

CBSE 2025

(A) Both assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the assertion (A).

16. *Assertion (A) :* The Balmer series in hydrogen atom spectrum is formed when the electron jumps from higher energy state to the ground state.

Reason (R) : In Bohr's model of hydrogen atom, the electron can jump between successive orbits only.

CBSE 2025

16. *Assertion (A) :* The Balmer series in hydrogen atom spectrum is formed when the electron jumps from higher energy state to the ground state.

Reason (R) : In Bohr's model of hydrogen atom, the electron can jump between successive orbits only.

CBSE 2025

(D) Both Assertion (A) and reason (R) are false

19. Define the term, 'distance of closest approach'. A proton of 3·95 MeV energy approaches a target nucleus Z = 79 in head-on position. Calculate its distance of closest approach. CBSE 2025 *2*

19. Define the term, 'distance of closest approach'. A proton of 3·95 MeV energy approaches a target nucleus Z = 79 in head-on position. Calculate its distance of closest approach. CBSE 2025 *2*

It is the distance from nucleus at which α particles stops momentarily and then begins to retrace its path.	
Alternatively	
It is the distance from nucleus at which entire intial kinetic energy of the α particle gets converted into electrostatic potential energy.	**1**
$r_0 = \frac{1}{4\pi\varepsilon_0}\frac{ze^2}{K.E}$	½
$= \frac{9\times10^9\times79\times(1.6\times10^{-19})^2}{3.95\times10^6\times1.6\times10^{-19}}$	
$= 28.8\times10^{-15}\ m$	½

20. In Bohr's model of hydrogen atom, find the percentage change in the radius of its orbit when an electron makes a transition from n = 3 state to n = 2 state. CBSE 2025 *2*

20. In Bohr's model of hydrogen atom, find the percentage change in the radius of its orbit when an electron makes a transition from n = 3 state to n = 2 state. CBSE 2025 *2*

$r \alpha n^2$	½
$\frac{r_2}{r_3} = \frac{4}{9}$	½
Percentage change when electron makes transition from n=3 to n=2	
$= \frac{\lvert r_2 - r_3 \rvert}{r_3} \times 100$	½
$= \left(\frac{\lvert 4 - 9 \rvert}{9} \right) \times 100$	
$= 55.55\%$	½

29. A hydrogen atom consists of an electron revolving in a circular orbit of radius r with certain velocity v around a proton located at the nucleus of the atom. The electrostatic force of attraction between the revolving electron and the proton provides the requisite centripetal force to keep it in the orbit. According to Bohr's model, an electron can revolve only in certain stable orbits. The angular momentum of the electron in these orbits is some integral multiple of $\frac{h}{2\pi}$, where h is the Planck's constant.

Further, when an electron makes a transition from one orbit of higher energy to that of lower energy, a photon is emitted having energy equal to the difference between energies of the initial and final states. Assuming the mass and charge of an electron as m and e respectively, answer the following questions.

CBSE 2025

(i) The expression for the speed of electron v in terms of radius of the orbit (r) and physical constant ($K = \frac{1}{4\pi\varepsilon_0}$) is : *1*

(A) $\frac{Ke^2}{mr}$ (B) $\frac{Ke^2}{mr^2}$

(C) $\sqrt{\frac{Ke^2}{mr}}$ (D) $\sqrt{\frac{Ke^2}{mr^2}}$

(i) The expression for the speed of electron v in terms of radius of the orbit (r) and physical constant ($K = \frac{1}{4\pi\varepsilon_0}$) is : *1*

(A) $\frac{Ke^2}{mr}$ (B) $\frac{Ke^2}{mr^2}$

(C) $\sqrt{\frac{Ke^2}{mr}}$ (D) $\sqrt{\frac{Ke^2}{mr^2}}$

(i) (C) $\sqrt{\frac{Ke^2}{mr}}$

(ii) The total energy of the atom in terms of r and physical constant K is : *1*

(A) $\frac{Ke^2}{r}$

(B) $-\frac{Ke^2}{2r}$

(C) $\frac{Ke^2}{2r}$

(D) $\frac{3}{2}\frac{Ke^2}{r}$

(ii) The total energy of the atom in terms of r and physical constant K is : *1*

(A) $\frac{Ke^2}{r}$

(B) $-\frac{Ke^2}{2r}$

(C) $\frac{Ke^2}{2r}$

(D) $\frac{3}{2}\frac{Ke^2}{r}$

(ii) (B) $\frac{-Ke^2}{2r}$

(iii) A photon of wavelength 500 nm is emitted when an electron makes a transition from one state to the other state in an atom. The change in the total energy of the electron and change in its kinetic energy in eV as per Bohr's model, respectively will be : *1*

(A) 2·48, – 2·48 (B) – 1·24, 1·24

(C) – 2·48, 2·48 (D) 1·24, – 1·24

(iii) A photon of wavelength 500 nm is emitted when an electron makes a transition from one state to the other state in an atom. The change in the total energy of the electron and change in its kinetic energy in eV as per Bohr's model, respectively will be : *1*

(A) 2·48, – 2·48 (B) – 1·24, 1·24

(C) – 2·48, 2·48 (D) 1·24, – 1·24

(iii) (C) -2.48, 2.48

(iv) (a) In Bohr's model of hydrogen atom, the frequency of revolution of electron in its n^{th} orbit is proportional to : *1*

(A) n

(B) $\frac{1}{n}$

(C) $\frac{1}{n^2}$

(D) $\frac{1}{n^3}$

(iv) (a) In Bohr's model of hydrogen atom, the frequency of revolution of electron in its n^{th} orbit is proportional to : *1*

(A) n

(B) $\frac{1}{n}$

(C) $\frac{1}{n^2}$

(D) $\frac{1}{n^3}$

(iv) (a)

(D) $\frac{1}{n^3}$

(b) An electron makes a transition from $-3{\cdot}4$ eV state to the ground state in hydrogen atom. Its radius of orbit changes by : (radius of orbit of electron in ground state = $0{\cdot}53$ Å) *1*

(A) $0{\cdot}53$ Å

(B) $1{\cdot}06$ Å

(C) $1{\cdot}59$ Å

(D) $2{\cdot}12$ Å

(b) An electron makes a transition from $-3{\cdot}4$ eV state to the ground state in hydrogen atom. Its radius of orbit changes by : (radius of orbit of electron in ground state = 0·53 Å) *1*

(A) 0·53 Å

(B) 1·06 Å

(C) 1·59 Å

(D) 2·12 Å

(C) 1.59 Å

20. Find the ratio of minimum to maximum wavelength of radiations emitted when electron jumps from higher energy state into ground state of hydrogen atom. *2*

20. Find the ratio of minimum to maximum wavelength of radiations emitted when electron jumps from higher energy state into ground state of hydrogen atom.

CBSE 2025 *2*

$$\frac{1}{\lambda}=R\left[\frac{1}{n_f^2}-\frac{1}{n_i^2}\right]$$

$$\frac{1}{\lambda_{max}}=R\left[\frac{1}{1^2}-\frac{1}{2^2}\right]$$

$$\lambda_{max}=\frac{4}{3R}$$

$$\frac{1}{\lambda_{min}}=R\left[\frac{1}{1^2}-\frac{1}{\infty}\right]$$

$$\lambda_{min}=\frac{1}{R}$$

$$\frac{\lambda_{min}}{\lambda_{max}}=\frac{3}{4}$$

Alternatively ½

for λ_{min}, $n_1=1$ $n_2=\infty$

$$E_2-E_1=\frac{hc}{\lambda_{min}}$$

$$0-(-13.6)=\frac{hc}{\lambda_{min}}$$ ½

$$\lambda_{min}=\frac{hc}{13.6}$$

λ_{max} $n_1=1$ $n_2=2$

$$\lambda_{max}=\frac{hc}{-3.4-(-13.6)}$$ ½

$$\lambda_{max}=\frac{hc}{10.2}$$

$$\frac{\lambda_{min}}{\lambda_{max}}=\frac{3}{4}$$ ½

11. Atomic spectral emission lines of hydrogen atom are incident on a zinc surface. The lines which can emit photoelectrons from the surface are members of **1**

(A) Balmer series
(B) Paschen series
(C) Lyman series
(D) Neither Balmer, nor Paschen nor Lyman series

CBSE 2025

11. Atomic spectral emission lines of hydrogen atom are incident on a zinc surface. The lines which can emit photoelectrons from the surface are members of **1**

(A) Balmer series
(B) Paschen series
(C) Lyman series
(D) Neither Balmer, nor Paschen nor Lyman series

CBSE 2025

(C) Lyman series

12. The energy of an electron in a hydrogen atom in ground state is –13.6 eV. Its energy in an orbit corresponding to quantum number n is –0.544 eV. The value of n is **1**

(A) 2
(B) 3
(C) 4
(D) 5

CBSE 2025

12. The energy of an electron in a hydrogen atom in ground state is –13.6 eV. Its energy in an orbit corresponding to quantum number n is –0.544 eV. The value of n is **1**

(A) 2 (B) 3

(C) 4 (D) 5

CBSE 2025

(D) 5

16. **Assertion (A)** : In Bohr model of hydrogen atom, the angular momentum of an electron in n^{th} orbit is proportional to the square root of its orbit radius r_n. **1**

Reason (R) : According to Bohr model, electron can jump to its nearest orbits only.

CBSE 2025

16. **Assertion (A)** : In Bohr model of hydrogen atom, the angular momentum of an electron in n^{th} orbit is proportional to the square root of its orbit radius r_n. **1**

Reason (R) : According to Bohr model, electron can jump to its nearest orbits only.

CBSE 2025

(C) If Assertion (A) is true but Reason (R) is false.

14. **Assertion (A)** : The potential energy of an electron revolving in any stationary orbit in a hydrogen atom is positive. **1**

Reason (R) : The total energy of a charged particle is always positive.

CBSE 2025

14. **Assertion (A)** : The potential energy of an electron revolving in any stationary orbit in a hydrogen atom is positive. **1**

Reason (R) : The total energy of a charged particle is always positive.

CBSE 2025

(D) Assertion (A) is false and Reason (R) is also false.

20. An electron in Bohr model of hydrogen atom makes a transition from energy level –1.51 eV to –3.40 eV. Calculate the change in the radius of its orbit. The radius of orbit of electron in its ground state is 0.53 Å. CBSE 2025 **2**

20. An electron in Bohr model of hydrogen atom makes a transition from energy level –1.51 eV to –3.40 eV. Calculate the change in the radius of its orbit. The radius of orbit of electron in its ground state is 0.53 Å. CBSE 2025 **2**

$E_n = \frac{-13.6}{n^2} eV$	½
$For\ E_n = -1.51 eV$	
$-1.51 = \frac{-13.6}{n^2}$	
n=3	½
$For\ E_n = -3.40 eV$	
$-3.40 = \frac{-13.6}{n^2}$	
n= 2	½
$\because r = 0.53 n^2\ \mathring{A}$	
$\therefore$ change in radius	
$\Delta r = 0.53[3^2 - 2^2]$	
$= 0.53 \times 5$	
$= 2.65\ \mathring{A}$	½

20. The energy of an electron in an orbit of Bohr hydrogen atom is -3.4 eV. Find its angular momentum. CBSE 2025 **2**

20. The energy of an electron in an orbit of Bohr hydrogen atom is –3.4 eV. Find its angular momentum. CBSE 2025 **2**

$E_n = -\frac{13.6}{n^2} eV$	½
$n^2 = \frac{-13.6}{-3.4} = 4$	½
n=2	
Angular momentum	
$L = \frac{nh}{2\pi}$	½
$L = \frac{h}{\pi} = \frac{6.63 \times 10^{-34}}{3.14} = 2.11 \times 10^{-34}$ Js	½

20. Prove that, in Bohr model of hydrogen atom, the time period of revolution of an electron in n^{th} orbit is proportional to n^3. CBSE 2025 **2**

20. Prove that, in Bohr model of hydrogen atom, the time period of revolution of an electron in n^{th} orbit is proportional to n^3. CBSE 2025 **2**

$T=\frac{2\pi r}{v}$ ---------- (1) ½

From Bohr's quantization condition

$mvr=\frac{nh}{2\pi}$

$v=\frac{nh}{2\pi mr}$ ---------- (2) ½

From (1) and (2)

$T=\frac{2\pi r}{\left(nh/2\pi mr\right)}$

$T=\frac{2\pi r(2\pi mr)}{nh}$

$T=\frac{4\pi^2 mr^2}{nh}$ ½

From $r=\frac{n^2h^2}{4\pi^2 mke^2}$

$T=\frac{4\pi^2 m}{nh}\left(\frac{n^2h^2}{4\pi^2 mke^2}\right)^2$

$T=\frac{n^3h^3}{4\pi^2 mk^2e^4}$

$\Rightarrow T\alpha n^3$ ½

Alternatively ---

$T=\frac{2\pi r}{v}$

$\because r\alpha n^2$

and $v\alpha\frac{1}{n}$

$\therefore T\alpha n^3$

9. Which of the following statements is correct for alpha particle scattering experiment ?

CBSE 2024 C

(A) For angle of scattering $\theta \approx 0$, the impact parameter is small.

(B) For angle of scattering $\theta \backsimeq \pi$, the impact parameter is large.

(C) The number of alpha particles undergoing head-on collision is small.

(D) The experiment provides an estimate of the upper limit to the size of target atom.

9. Which of the following statements is correct for alpha particle scattering experiment ?

CBSE 2024 C

(A) For angle of scattering $\theta \approx 0$, the impact parameter is small.

(B) For angle of scattering $\theta \simeq \pi$, the impact parameter is large.

(C) The number of alpha particles undergoing head-on collision is small.

(D) The experiment provides an estimate of the upper limit to the size of target atom.

(C) The number of alpha particles undergoing head on collision is small.

(i) State Bohr's postulates and derive an expression for the energy of electron in n^{th} orbit in Bohr's model of hydrogen atom. CBSE 2024 C

(i) State Bohr's postulates and derive an expression for the energy of electron in n^{th} orbit in Bohr's model of hydrogen atom. CBSE 2024 C

(i)

Bohr's Postulates

(a) Bohr's first postulate was that an electron in an atom could revolve in certain stable orbits without the emission of radiant energy, ½

(b) Bohr's second postulate defines these stable orbits. This postulate states that the electron revolves around the nucleus only in those orbits for which the angular momentum is some integral multiple of $h/2\pi$ where h is the Planck's constant. ½

(c) Bohr's third postulate states that an electron might make a transition from one of its specified non-radiating orbits to another of lower energy. When it does so, a photon is emitted having energy equal to the energy difference between the initial and final states. ½

Derivation

Total energy of electron in the stationary state of hydrogen atoms is

$$E = -\frac{e^2}{8\pi\varepsilon_0 r_n} \text{ ----------(1)}$$ ½

Where r_n is radius of n^{th} orbit

$$r_n = \frac{n^2 h^2 \varepsilon_0}{\pi m e^2} \text{ ----------(2)}$$ ½

Substituting eq (2) in eq (1)

$$E_n = -\frac{me^4}{8n^2\varepsilon_0{}^2 h^2}$$ 1

9. The ionisation energy of the hydrogen atom, in Bohr model, is :

(A) $-3{\cdot}4$ eV

(B) $3{\cdot}4$ eV

(C) $-13{\cdot}6$ eV

(D) $13{\cdot}6$ eV

CBSE 2024 C

9. The ionisation energy of the hydrogen atom, in Bohr model, is :

(A) – 3·4 eV

(B) 3·4 eV

(C) – 13·6 eV

(D) 13·6 eV

CBSE 2024 C

(D) 13.6 e V

10. An electron makes a transition from orbit n = 2 to orbit n = 1, in Bohr's model of hydrogen atom. Consider change in magnitudes of its kinetic energy (K) and potential energy (U).

(A) K increases and U decreases

(B) K decreases and U increases

(C) Both K and U decrease

(D) Both K and U increase

CBSE 2024 C

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(B) K decreases and U increases

(C) Both K and U decrease

(D) Both K and U increase

CBSE 2024 C

(A) K increases and U decreases

10. The electron of a hydrogen atom is in the second excited level. During its de-excitation to the ground state, the number of possible spectral lines that can be emitted is :

(A) 1 (B) 2

(C) 3 (D) 6

CBSE 2024 C

10. The electron of a hydrogen atom is in the second excited level. During its de-excitation to the ground state, the number of possible spectral lines that can be emitted is :

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(C) 3 (D) 6

CBSE 2024 C

(C) 3

20. What are the postulates of Bohr's theory of hydrogen atom ? Write the expression of radius of n^{th} orbit of electron. CBSE 2024 C *2*

20. What are the postulates of Bohr's theory of hydrogen atom ? Write the expression of radius of n^{th} orbit of electron. CBSE 2024 C *2*

(i) Bohr's first postulate states that an electron in an atom revolve in certain stable orbits without the emission of radiant energy.	½
(ii) Bohr's second postulate states that the electron revolves around the nucleus only in those orbits for which the angular momentum is some integral multiple of $h/2\pi$ where h is the Planck's constant (= 6.6×10^{-34} J s).Thus the angular momentum (L) of the orbiting electron is quantized. That is $L = nh/2\pi$	½
(iii) Bohr's third postulate states that an electron might make a transition from one of its specified non-radiating orbits to another of lower energy. When it does so, a photon is emitted having energy equal to the energy difference between the initial and final states.	½
Radius of n^{th} orbit of electron, $r_n = \frac{n^2h^2\varepsilon_o}{\pi me^2}$	½

30. When an electron of hydrogen atom jumps from a ground state energy level to higher energy level, it absorbs energy and is excited. It can remain in this state for a maximum time $\simeq 10^{-7}$s. When de-excited, the electron returns back and the same amount of energy is radiated in the form of a photon. The difference of energies of two energy levels ($E_f - E_i$) is the energy of this photon. Depending upon the two levels involved in transition, the spectral lines are grouped in five series of hydrogen line emission spectrum. These are named as Lyman ($n_f = 1$), Balmer ($n_f = 2$), Paschen ($n_f = 3$), Brackett ($n_f = 4$) and Pfund ($n_f = 5$) series. The values of energy of the electron in the n^{th} level of hydrogen atom and of hydrogen-like atom are $\frac{-13.6}{n^2}$ eV and $\frac{-13.6}{n^2} Z^2$ eV respectively. Here Z is number of protons in the nucleus of hydrogen-like atom.

CBSE 2024 C

(i) Which one of the following hydrogen spectral series lie in visible region ? *1*

(A) Lyman
(B) Balmer
(C) Pfund
(D) Paschen

CBSE 2024 C

(i) Which one of the following hydrogen spectral series lie in visible region ? *1*

(A) Lyman (B) Balmer

(C) Pfund (D) Paschen

CBSE 2024 C

(B) Balmer

(ii) In a hydrogen atom, when an electron jumps from n = 6 to n = 3 level, the spectral line obtained belongs to : *1*

(A) Lyman (B) Pfund

(C) Brackett (D) Paschen

CBSE 2024 C

(ii) In a hydrogen atom, when an electron jumps from n = 6 to n = 3 level, the spectral line obtained belongs to : *1*

(A) Lyman (B) Pfund

(C) Brackett (D) Paschen

CBSE 2024 C

(D) Pascher

(iii) The second member of Lyman spectral series in absorption hydrogen spectrum results due to electron transition : *1*

(A) $n = 3$ to $n = 1$ (B) $n = 1$ to $n = 3$

(C) $n = 1$ to $n = 2$ (D) $n = 2$ to $n = 1$

CBSE 2024 C

(iii) The second member of Lyman spectral series in absorption hydrogen spectrum results due to electron transition : *1*

(A) $n = 3$ to $n = 1$ (B) $n = 1$ to $n = 3$

(C) $n = 1$ to $n = 2$ (D) $n = 2$ to $n = 1$

CBSE 2024 C

(B) $n = 1$ to $n = 3$

(iv) (a) The ratio of energies in Bohr hydrogen atom and in hydrogen-like He atom for the first electron orbit is : *1*

(A) 1 : 4 (B) 4 : 1

(C) 2 : 1 (D) 1 : 2

CBSE 2024 C

(iv) (a) The ratio of energies in Bohr hydrogen atom and in hydrogen-like He atom for the first electron orbit is : *1*

(A) 1 : 4 (B) 4 : 1

(C) 2 : 1 (D) 1 : 2

CBSE 2024 C

(a) (A) 1:4

(iv) (b) The spacing between successive energy levels of hydrogen atom : *1*

(A) is same for all levels

(B) becomes larger for higher levels

(C) becomes smaller for higher levels

(D) may be larger or smaller for higher levels depending upon temperature

CBSE 2024 C

(iv) (b) The spacing between successive energy levels of hydrogen atom : *1*

(A) is same for all levels

(B) becomes larger for higher levels

(C) becomes smaller for higher levels

(D) may be larger or smaller for higher levels depending upon temperature

CBSE 2024 C

(C) becomes smaller for higher levels

8. The energy of an electron in the ground state of hydrogen atom is –13.6 eV. The kinetic and potential energy of the electron in the first excited state will be

(A) –13.6 eV, 27.2 eV
(B) –6.8 eV, 13.6 eV
(C) 3.4 eV, –6.8 eV
(D) 6.8 eV, –3.4 eV

CBSE 2024

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(C) 3.4 eV, –6.8 eV
(D) 6.8 eV, –3.4 eV

CBSE 2024

(C) $3.4eV, -6.8eV$

14. **Assertion (A)** : An alpha particle is moving towards a gold nucleus. The impact parameter is maximum for the scattering angle of 180°.

Reason (R) : The impact parameter in an alpha particle scattering experiment does not depend upon the atomic number of the target nucleus.

CBSE 2024

14. **Assertion (A)** : An alpha particle is moving towards a gold nucleus. The impact parameter is maximum for the scattering angle of 180°.

Reason (R) : The impact parameter in an alpha particle scattering experiment does not depend upon the atomic number of the target nucleus.

CBSE 2024

(D) Both Assertion (A) and Reason (R) are false.

21. The electron in hydrogen atom is revolving with the speed of 2.2×10^6 m/s in an orbit of radius 0.53 Å. Calculate the initial frequency of light emitted by the electron using classical physics.

CBSE 2024

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CBSE 2024

$\nu = \frac{v}{2\pi r}$	1
$\nu = \frac{2.2 \times 10^6}{2 \times \pi \times 0.53 \times 10^{-10}}$	½
$\nu = 6.6 \times 10^{15}\,\text{Hz}$	½

21. A proton of energy 1.6 MeV approaches a gold nucleus (Z = 79). Find the distance of its closest approach.

CBSE 2024

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CBSE 2024

$d_0 = \frac{kZe^2}{K_p}$	½
$= \frac{9\times10^9 \times 79 \times (1.6\times10^{-19})^2}{1.6\times1.6\times10^{-19}\times10^6}$	½ + ½
$= 711\times10^{-16}\,\text{m}$	½
$= 7.11\times10^{-14}\,\text{m}$	

20. (a) Two energy levels of an electron in hydrogen atom are separated by 2.55 eV. Find the wavelength of radiation emitted when the electron makes transition from the higher energy level to the lower energy level.

(b) In which series of hydrogen spectrum this line shall fall ? CBSE 2024

20. (a) Two energy levels of an electron in hydrogen atom are separated by 2.55 eV. Find the wavelength of radiation emitted when the electron makes transition from the higher energy level to the lower energy level.

(b) In which series of hydrogen spectrum this line shall fall ? CBSE 2024

(a) $E_2 - E_1 = \frac{hc}{\lambda}$	½
Given	
$E_2 - E_1 = 2.55 \times 1.6 \times 10^{-19}$ J	½
$\Rightarrow \quad \lambda = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{2.55 \times 1.6 \times 10^{-19}} = 487.5\,\text{nm}$	½
(b) Balmer series	½

4. In Balmer series of hydrogen atom, as the wavelength of spectral lines decreases, they appear **1**

(A) equally spaced and equally intense.

(B) further apart and stronger in intensity.

(C) closer together and stronger in intensity.

(D) closer together and weaker in intensity.

CBSE 2024

4. In Balmer series of hydrogen atom, as the wavelength of spectral lines decreases, they appear **1**

(A) equally spaced and equally intense.

(B) further apart and stronger in intensity.

(C) closer together and stronger in intensity.

(D) closer together and weaker in intensity.

CBSE 2024

(D) Closer together and weaker in intensity.

24. Draw the graph showing variation of scattered particles detected (N) with the scattering angle (θ) in Geiger-Marsden experiment. Write two conclusions that you can draw from this graph. Obtain the expression for the distance of closest approach in this experiment. CBSE 2024 **3**

24. Draw the graph showing variation of scattered particles detected (N) with the scattering angle (θ) in Geiger-Marsden experiment. Write two conclusions that you can draw from this graph. Obtain the expression for the distance of closest approach in this experiment. CBSE 2024 **3**

Two conclusions

(i) Most of an atom is empty space. ½

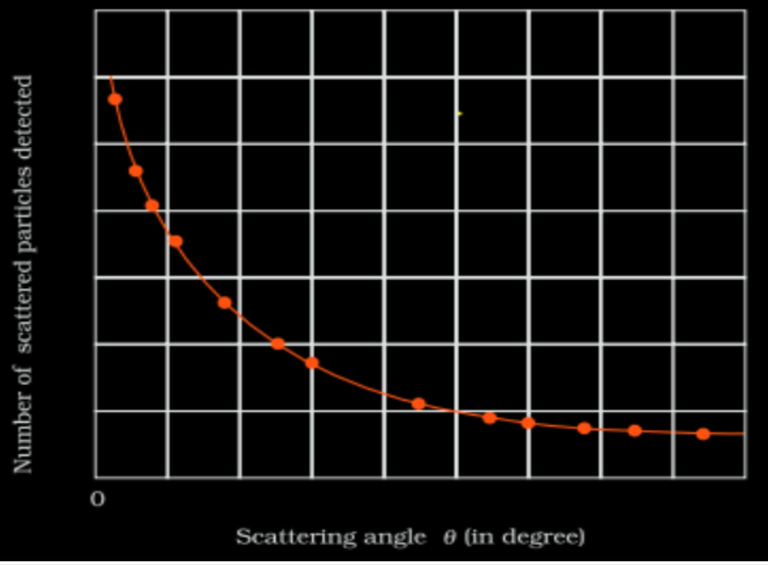
(ii) Almost entire mass and entire positive charge is concentrated in a very small region called nucleus. ½

At distance of closest approach ½

$$E_k = E_P$$

$$K = \frac{1}{4\pi\varepsilon_0}\frac{(Ze).(2e)}{d}$$

$$d = \frac{1}{4\pi\varepsilon_0}\frac{(2Ze^2)}{K}$$



½

8. The radius (r_n) of n^{th} orbit in Bohr model of hydrogen atom varies with n as **1**

(A) $r_n \propto n$

(B) $r_n \propto \frac{1}{n}$

(C) $r_n \propto n^2$

(D) $r_n \propto \frac{1}{n^2}$

CBSE 2024

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(D) $r_n \propto \frac{1}{n^2}$

CBSE 2024

(C) $r_n \alpha n^2$

9. A proton and an alpha particle having equal velocities approach a target nucleus. They come momentarily to rest and then reverse their directions. The ratio of the distance of closest approach of the proton to that of the alpha particle will be :

(A) $\frac{1}{2}$ (B) 2

(C) $\frac{1}{4}$ (D) 4

CBSE 2024

9. A proton and an alpha particle having equal velocities approach a target nucleus. They come momentarily to rest and then reverse their directions. The ratio of the distance of closest approach of the proton to that of the alpha particle will be :

(A) $\frac{1}{2}$ (B) 2

(C) $\frac{1}{4}$ (D) 4

CBSE 2024

(B) 2

11. An electron makes a transition from n = 2 level to n = 1 level in the Bohr model of a hydrogen atom. Its period of revolution :

(A) increases by 87·5%

(B) decreases by 87·5%

(C) increases by 43·75%

(D) decreases by 43·75%

CBSE 2024

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(B) decreases by 87·5%

(C) increases by 43·75%

(D) decreases by 43·75%

CBSE 2024

(B) decreased by 87.5%

26. State Bohr's second postulate for his theory of hydrogen atom. Prove that the radius of Bohr's orbit of a hydrogen atom is directly proportional to n^2, where n is the principal quantum number. CBSE 2024 *3*

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Bohr's second postulate Electron revolves around the nucleus only in those orbits for which the angular momentum in some integral multiple of $h/2\pi$. Electrostatic force between revolving electron & nucleus provide requisite centripetal force	½
$\frac{mv_n^2}{r_n} = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r_n^2}$	½
$v_n = \frac{e}{\sqrt{4\pi\varepsilon_0 m r_n}}$ _____(i)	½
$mv_n r_n = \frac{nh}{2\pi}$ _____(ii)	½
From eqn. (i) and (ii)	
$r_n = \left(\frac{n^2}{m}\right)\left(\frac{h}{2\pi}\right)^2 \frac{4\pi\varepsilon_0}{e^2}$	½
$r_n \propto n^2$	½

26. Explain the origin of the spectral lines of hydrogen atom using Bohr's theory. Draw the energy level diagram of hydrogen atom showing its various spectral series. CBSE 2024 *3*

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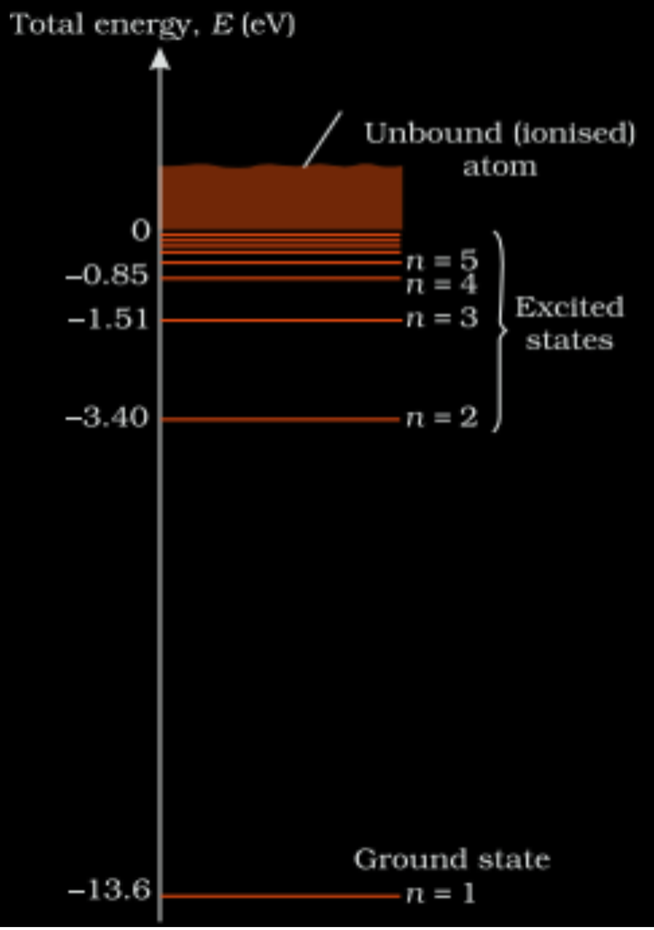
When an electron makes a transition from higher energy level to a lower energy orbit, a photon is emitted having energy equal to energy difference between these two orbits. **1**



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26. State Bohr's first and second postulates. Use them to derive an expression for the radius of the n^{th} orbit in a hydrogen atom. CBSE 2024 *3*

26. State Bohr's first and second postulates. Use them to derive an expression for the radius of the n^{th} orbit in a hydrogen atom. CBSE 2024 *3*

- **Bohr's first postulate**
  An electron in an atom revolves in certain stable orbits without the emission of radiant energy. ½
- **Bohr's second postulate**
  Electron revolves around the nucleus only in those orbits for which the angular momentum is integral multiple of $\frac{h}{2\pi}$. ½

Electrostatic force between revolving electron and nucleus provides requisite centripetal force

$$\frac{mv_n^2}{r_n} = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r_n^2}$$ ½

$$v_n = \frac{e}{\sqrt{4\pi\varepsilon_0 m r_n}} \qquad ------(i)$$ ½

$$mv_n r_n = \frac{nh}{2\pi} \qquad ------(ii)$$ ½

using equations (i) and (ii)

$$r_n = \left(\frac{n^2}{m}\right)\left(\frac{h}{2\pi}\right)^2 \frac{4\pi\varepsilon_0}{e^2}$$ ½

9. Energy levels A, B and C of an atom correspond to increasing values of energy i.e. $E_A < E_B < E_C$. Let λ_1, λ_2 and λ_3 be the wavelengths of radiation corresponding to the transitions C to B, B to A and C to A, respectively. The correct relation between λ_1, λ_2 and λ_3 is :

(A) $\lambda_1^2 + \lambda_2^2 = \lambda_3^2$

(B) $\frac{1}{\lambda_1} + \frac{1}{\lambda_2} = \frac{1}{\lambda_3}$

(C) $\lambda_1 + \lambda_2 + \lambda_3 = 0$

(D) $\lambda_1 + \lambda_2 = \lambda_3$

CBSE 2024

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(C) $\lambda_1 + \lambda_2 + \lambda_3 = 0$

(D) $\lambda_1 + \lambda_2 = \lambda_3$

CBSE 2024

(B) $\frac{1}{\lambda_1} + \frac{1}{\lambda_2} = \frac{1}{\lambda_3}$

10. An alpha particle approaches a gold nucleus in Geiger-Marsden experiment with kinetic energy K. It momentarily stops at a distance d from the nucleus and reverses its direction. Then d is proportional to :

(A) $\frac{1}{\sqrt{K}}$

(B) $\sqrt{K}$

(C) $\frac{1}{K}$

(D) K

CBSE 2024

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(A) $\frac{1}{\sqrt{K}}$ (B) $\sqrt{K}$

(C) $\frac{1}{K}$ (D) K

CBSE 2024

(C) $\frac{1}{K}$

26. Using Bohr's postulates, derive the expression for the radius of the n^{th} orbit of an electron in a hydrogen atom. Also find the numerical value of Bohr's radius a_0. CBSE 2024 *3*

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CBSE 2024 *3*

From Bohr's second postulate

$$m v r = \frac{nh}{2\pi} \qquad \ldots\ldots(1)$$ ½

Also $\frac{mv^2}{r} = \frac{e^2}{4\pi\varepsilon_0 r^2}$ (z=1) ½

$$v = \frac{e}{\sqrt{4\pi\varepsilon_0 mr}}$$ ½

Substituting in (1) and simplifying

$$r = \frac{n^2 h^2 \varepsilon_0}{\pi m e^2}$$ ½

For n = 1 r = a_o (Bohr's radius)

$$a_o = \frac{(6.63\times10^{-34})^2 \times 8.854\times10^{-12}}{3.14\times9.1\times10^{-31}\times(1.6\times10^{-19})^2}$$ ½

$= 5.29\text{x}10^{-11}\text{m}$

$= 0.53\text{Å}$ ½

27. Draw a schematic arrangement of Geiger-Marsden experiment. Briefly explain the conclusions drawn from the observations about the structure of an atom. Define 'distance of closest approach' in this case. CBSE 2024 *3*

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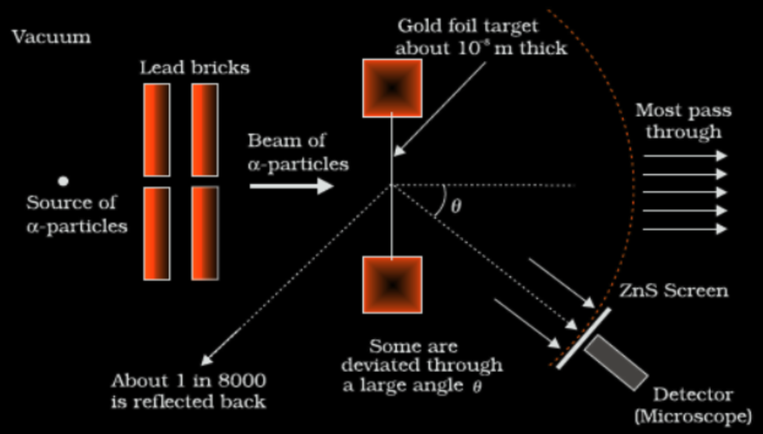

1

(b) -Entire positive charge and most of the mass of atom are concentrated in the nucleus.
- Electrons move in orbits about the nucleus just as planets around the sun. **1**
- Size of nucleus is about 10^{-15}m to 10^{-14}m.
- Most of the space in an atom is empty.

(c) The distance of the α particle from the centre of nucleus at which the whole of the initial kinetic energy of the particle gets converted into the electric potential energy. **1**

Alternatively

The distance of the α particle from the centre of nucleus at which it stops momentarily and reverses its direction.

11. The transition of electron that gives rise to the formation of the second spectral line of the Balmer series in the spectrum of hydrogen atom corresponds to :

(A) $n_f = 2$ and $n_i = 3$

(B) $n_f = 3$ and $n_i = 4$

(C) $n_f = 2$ and $n_i = 4$

(D) $n_f = 2$ and $n_i = \infty$

CBSE 2024

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(C) $n_f = 2$ and $n_i = 4$

(D) $n_f = 2$ and $n_i = \infty$

CBSE 2024

(C) $n_f = 2$ and $n_i = 4$

27. State the three postulates of Bohr's theory of hydrogen atom.

A hydrogen atom de-excites from level n to level (n – 1). Show that, according to Bohr's theory, the frequency of radiation emitted $\nu \cong \frac{\alpha}{n^3}$, for large values of n, where α is a constant. This result exactly agrees with that obtained from classical physics – one of the successes of Bohr's theory.

CBSE 2024 *3*

Bohr's postulates:-

(i) An electron in an atom could revolve in certain stable orbits without the emission of radiant energy.

(ii) The electron revolves around the nucleus only in those orbits for which the angular momentum is some integral multiple of $h/2\pi$

(iii) An electron might make a transition from one of its specified non-radiating orbits to another of lower energy. When it does so, a photon is emitted having energy equal to the energy difference between the initial and final states. ½ + ½ +½

$h\nu = E_i - E_f$ ------- (i)

$E_n \propto \frac{1}{n^2}$ ½

$$E_n - E_{n-1} = k\left[\frac{1}{(n-1)^2} - \frac{1}{n^2}\right]$$

$$= k\frac{n^2 - \left[n^2 + 1 - 2n\right]}{n^4 - 2n^3 + n^2}$$

$$= k\frac{2n-1}{n^4 - 2n^3 + n^2}$$

For large n , $2n-1 \cong 2n$

$n^2 [n^2 - 2n+1] \cong n^4$

$$E_n - E_{n-1} \cong k\frac{2n}{n^4} \cong \frac{2k}{n^3}$$ ----- (ii) 1

From equation (i) and (ii)

$\nu \cong \frac{\alpha}{n^3}$

26. Write the drawbacks of Rutherford's atomic model. How did Bohr remove them ? Show that different orbits in Bohr's atom are not equally spaced. *3* CBSE 2024

26. Write the drawbacks of Rutherford's atomic model. How did Bohr remove them ? Show that different orbits in Bohr's atom are not equally spaced. *3* CBSE 2024

Drawbacks: i) According to classical electromagnetic theory, an accelerating charged particle emits radiation in the form of electromagnetic waves. The energy of an accelerating electron should therefore, continuously decrease. The electron would spiral inward and eventually fall into the nucleus. Thus, such an atom cannot be stable. ii) As the electrons spiral inwards, their angular velocities and hence their frequencies would change continuously. Thus, they would emit a continuous spectrum, in contradiction to the line spectrum actually observed.	1
Bohr postulated stable orbits in which electrons do not radiate energy	1
Alternatively: Bohr's postulates (Any ONE of the three) (i) An electron in an atom could revolve in certain stable orbits without the emission of radiant energy. (ii) The electron revolves around the nucleus only in those orbits for which the angular momentum is some integral multiple of $h/2\pi$ (iii) An electron might make a transition from one of its specified non-radiating orbits to another of lower energy. When it does so, a photon is emitted having energy equal to the energy difference between the initial and final states.	

9. In Bohr model of a hydrogen atom, the increase in the angular momentum of the electron, when it is excited from n = 1 to n = 3 orbit, is :

(A) $\frac{h}{2\pi}$

(B) $\frac{2h}{\pi}$

(C) $\frac{3h}{2\pi}$

(D) $\frac{h}{\pi}$

CBSE 2024

9. In Bohr model of a hydrogen atom, the increase in the angular momentum of the electron, when it is excited from n = 1 to n = 3 orbit, is :

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(B) $\frac{2h}{\pi}$

(C) $\frac{3h}{2\pi}$

(D) $\frac{h}{\pi}$

CBSE 2024

(D) h/π

10. A positively charged particle of mass m, moving with velocity v is bombarded on a heavy nucleus of charge Ze. Its distance of closest approach from the nucleus is proportional to :

(A) $\frac{Z}{mv^2}$

(B) $\frac{v^2}{Zm}$

(C) v^2

(D) $\frac{Zv^2}{m}$

CBSE 2024

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(B) $\frac{v^2}{Zm}$

(C) v^2

(D) $\frac{Zv^2}{m}$

CBSE 2024

(A) Z/mv^2

27. Explain the origin of Lyman and Balmer series in the emission spectrum of hydrogen atom. How are the maximum and the minimum wavelength of lines in a series determined ? CBSE 2024 *3*

27. Explain the origin of Lyman and Balmer series in the emission spectrum of hydrogen atom. How are the maximum and the minimum wavelength of lines in a series determined ? CBSE 2024 *3*

In Hydrogen Spectrum, Lyman series is originated when an electron jumps from $n \geq 2$ to $n=1$.	½
In Hydrogen Spectrum, Balmer series is originated when an electron jumps from $n \geq 3$ to $n=2$.	½
Wavelength λ, $\frac{1}{\lambda} = R\left[\frac{1}{n_f^2} - \frac{1}{n_i^2}\right]$	
<u>Lyman series</u> The maximum wavelength for Lyman series is determined when an electron jumps from $n_i = 2$ to $n_f = 1$ and minimum wavelength is determined when an electron jumps from $n_i = \infty$ to $n_f = 1$.	½+ ½
<u>Balmer series</u> The maximum wavelength for Balmer series is determined when an electron jumps from $n_i = 3$ to $n_f = 2$ and minimum wavelength is determined when an electron jumps from $n_i = \infty$ to $n_f = 2$.	½+ ½

11. The minimum energy required to free the electron from the ground state of the hydrogen atom in Bohr model is 13·6 eV. The energy of the electron in its second orbit will be :

(a) – 6·8 eV (b) 6·8 eV

(c) – 3·4 eV (d) 3·4 eV

CBSE 2023 C

11. The minimum energy required to free the electron from the ground state of the hydrogen atom in Bohr model is 13·6 eV. The energy of the electron in its second orbit will be :

(a) – 6·8 eV (b) 6·8 eV

(c) – 3·4 eV (d) 3·4 eV

CBSE 2023 C

(c) -3.4 eV

12. Which of the following series in hydrogen spectrum lies in its visible region ?

(a) Lyman series

(b) Balmer series

(c) Paschen series

(d) Pfund series

CBSE 2023 C

12. Which of the following series in hydrogen spectrum lies in its visible region ?

(a) Lyman series

(b) Balmer series

(c) Paschen series

(d) Pfund series

CBSE 2023 C

(b) Balmer series

(b) An electron (charge – e, mass m) is revolving around a positively charged (charge + e) nucleus in a circular orbit of radius r. *3*

(i) For a stable orbit, equating the electrostatic Coulomb force between the electron and the nucleus with the requisite centripetal force, obtain an expression for radius r.

(ii) What is Bohr's second postulate about angular momentum of electron in its orbit ? Using it and expression obtained in (i) above, show that the radius r of the n^{th} possible orbit is directly proportional to n^2.

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(i) Electrostatic force between the electron and the nucleus.

$$F = \frac{ke^2}{r^2}$$ ½

At it provides centripetal force

$$\frac{ke^2}{r^2} = \frac{mv^2}{r}$$

$$r = \frac{ke^2}{mv^2}$$

(ii) "Electrons revolves around the nucleus only in those orbits for which the angular momentum is some integral multiple of $\frac{h}{2\pi}$" ½ ½

$$mvr = \frac{nh}{2\pi}$$

$$v = \frac{nh}{2\pi mr} \text{ _____ (2)}$$ ½

Substituting value of 'v' from equation (2) in equation(1)

$$r = \frac{ke^2h}{mn^2h^2} \times 4\pi^2m^2r^2$$ ½

$$r = \frac{n^2h^2}{4\pi^2mke^2}$$

$$r \propto n^2$$ ½

12. The impact parameter for an alpha particle approaching a target nucleus is maximum when the scattering angle (θ) is :

(a) $0°$ (b) $90°$

(c) $180°$ (d) $45°$

CBSE 2023 C

12. The impact parameter for an alpha particle approaching a target nucleus is maximum when the scattering angle (θ) is :

(a) $0°$ (b) $90°$

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CBSE 2023 C

(a) 0^o

27. (a) (i) An electron in a hydrogen atom jumps from second excited state to the first excited state. Name the spectral series in the spectrum of hydrogen atom to which the emitted radiation belongs.

(ii) Find the ratio of the wavelengths of the "most energetic spectral" lines in the Balmer series to that in Paschen series of Hydrogen spectrum. CBSE 2023 C *3*

27. (a) (i) An electron in a hydrogen atom jumps from second excited state to the first excited state. Name the spectral series in the spectrum of hydrogen atom to which the emitted radiation belongs.

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CBSE 2023 C

3

(a) (i) Balmer series	½
(ii) $\frac{1}{\lambda_B} = R[\frac{1}{2^2} - \frac{1}{\infty}]$	½
$\frac{1}{\lambda_B} = \frac{R}{4}$	½
$\frac{1}{\lambda_p} = R[\frac{1}{3^2} - \frac{1}{\infty}]$	½
$\frac{1}{\lambda p} = \frac{R}{9}$	½
$\frac{\lambda_B}{\lambda p} = \frac{4}{9}$	½

(b) (i) An α-particle having kinetic energy K approaches a nucleus of atomic number Z. It gets close to the nucleus and then approaches a distance (d) and reverses its direction. Obtain an expression for the distance of closest approach (d) in terms of kinetic energy of the α-particle.

CBSE 2023 C

(ii) A proton and an alpha particle approach a target nucleus in head-on position, with equal velocities. Find the ratio of their distances of closest approach to the target nucleus. *3*

(i) At the distance of closest approach.	
$K = \frac{(Ze)2e}{4\pi\varepsilon_o d}$	½
$d = \frac{2Ze^2}{4\pi\varepsilon_o K}$	1
$d = \frac{2(Ze)q}{\frac{1}{2}mv^2 \times 4\pi\varepsilon_o}$	½
$d = \frac{(Ze)q}{mv^2\pi\varepsilon_o}$	
$d\alpha \frac{q}{m}$	½
$\frac{d_p}{d_\alpha} = \frac{e}{2e} \times \frac{4m}{m} = \frac{2}{1}$	½

12. A photon absorbed by a hydrogen atom excites it from n = 3 level to n = 5 level. The energy of the photon is :

(a) +0·97 eV (b) – 0·97 eV

(c) +1·51 eV (d) – 0·54 eV

CBSE 2023

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(a) +0·97 eV (b) – 0·97 eV

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CBSE 2023

(a) + 0.97eV

23. State Bohr's postulates of hydrogen atom. How is a photon emitted by an excited hydrogen atom ? CBSE 2023 *2*

23. State Bohr's postulates of hydrogen atom. How is a photon emitted by an excited hydrogen atom ? CBSE 2023 *2*

(i)An electron in an atom revolve in certain stable orbit without emission of radiant energy. (ii) The electron revolves around the nucleus only in those orbits for which the angular momentum is some integral multiple of $\frac{h}{2\pi}$. i.e. $L = nh/2\pi$ (iii)When an electron jumps from excited state of energy E_2 to the lower state of energy E_1 photon having energy $h\nu = E_2 - E_1$ is emitted. When electron jumps from higher energy level to lower energy level the energy is released in the form of photon is emitted.	1½ ½

(b) Briefly explain the observations made in Geiger-Marsden experiment. Write the important conclusions drawn about the structure of atom from these observations. *3*

CBSE 2023

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CBSE 2023

Observations: (i) Most of the α-particle passed straight. (ii) A few α-particles scattered at small angle and very few scattered at an angle $\theta > 90^o$ Conclusions: (i) The positive charge and the mass of the atom are concentrated at the nucleus of the atom. (ii) Scattering of α- particle was due to the Coulomb repulsive force between the positively charged α- particle and the nucleus.	½ ½ 1 1

14. The potential energy of an electron in the second excited state in hydrogen atom is :

CBSE 2023

(a) $-3{\cdot}4$ eV (b) $-3{\cdot}02$ eV (c) $-1{\cdot}51$ eV (d) $-6{\cdot}8$ eV

14. The potential energy of an electron in the second excited state in hydrogen atom is :

CBSE 2023

(a) $-3{\cdot}4$ eV (b) $-3{\cdot}02$ eV (c) $-1{\cdot}51$ eV (d) $-6{\cdot}8$ eV

(b) – 3.02 eV

28. (a) Differentiate between 'impact parameter' and 'distance of closest approach' in alpha particle scattering experiment. Identify, situations when impact parameter is (i) minimum, and (ii) very large.

CBSE 2023

(b) An electron cannot revolve around the nucleus in a hydrogen atom at an arbitrary distance. Explain. *3*

28. (a) Differentiate between 'impact parameter' and 'distance of closest approach' in alpha particle scattering experiment. Identify, situations when impact parameter is (i) minimum, and (ii) very large.

CBSE 2023

(b) An electron cannot revolve around the nucleus in a hydrogen atom at an arbitrary distance. Explain. *3*

(a) Impact parameter is the perpendicular distance of the initial velocity vector of the α – particle from the centre of the nucleus.	½
Distance of closest approach**:** It is the minimum distance between the projected α – particle and the nucleus of target atom at which the kinetic energy of the α – particle becomes equal to potential energy of α – particle in the field of nucleus.	½
Minimum: When α – particle rebounds back ($\theta = \pi$)	½
Very large: when α – particle goes nearly undeviated and has a small deflection ($\theta = 0^\circ$)	½
(b) According to Bohr's second postulate electron revolves only in those orbit for which angular momentum is $\frac{nh}{2\pi}$. The observed spectral lines of hydrogen atom confirms this.	1

30. Briefly explain Geiger-Marsden experiment. Show the variation of the number of particles scattered (N) with scattering angle (θ) in this experiment. What is the main conclusion that can be inferred from this plot ?

CBSE 2023 *3*

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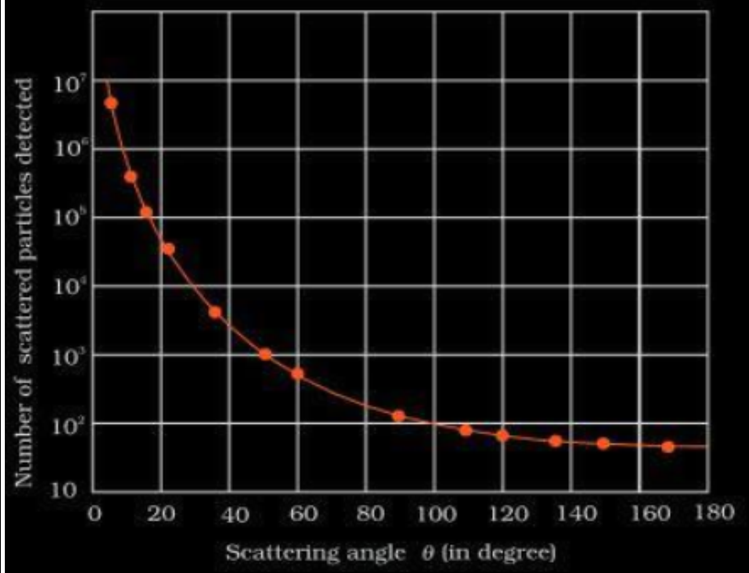
CBSE 2023 *3*



Alpha-particles emitted by a radioactive source were allowed to fall on a thin foil of gold .The scattered alpha-particles were observed through a rotatable detector consisting of zinc sulphide screen and a microscope. The scattered alpha-particles on striking the screen produced brief light flashes or scintillations. 1

Conclusion:

The existence of positively charged nucleus inside an atom and provide an upper limit to the size of the nucleus. 1

1

13. The ratio of maximum frequency and minimum frequency of light emitted in Balmer series of hydrogen spectrum, in Bohr's model is :

(a) $\frac{11}{9}$

(b) $\frac{9}{5}$

(c) $\frac{11}{7}$

(d) $\frac{16}{7}$

CBSE 2023

13. The ratio of maximum frequency and minimum frequency of light emitted in Balmer series of hydrogen spectrum, in Bohr's model is :

(a) $\frac{11}{9}$

(b) $\frac{9}{5}$

(c) $\frac{11}{7}$

(d) $\frac{16}{7}$

CBSE 2023

(b) $\frac{9}{5}$

(b) Using Bohr's postulates, obtain the expression for radius of n^{th} stable orbit in a hydrogen atom. CBSE 2023 *2*

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$\frac{mv^2}{r_n} = \frac{kq^2}{r_n^2} = \frac{e^2}{4\pi \in_o r_n^2}$ -----(1)	½
$mvr_n = \frac{nh}{2\pi}$ -----(2)	½
Using equation (1) &(2)	
$r_n = \frac{n^2 h^2 4\pi\varepsilon_o}{m(2\pi)^2 e^2} = 0.53 \times 10^{-10} n^2$ m	1

(b) (i) Draw a graph to show the variation of the number of scattered particles detected (N) in Geiger-Marsden experiment as a function of scattering angle (θ).

(ii) Discuss briefly two conclusions that can be drawn from this graph and how they lead to the discovery of nucleus in an atom.

CBSE 2023

3

(b) (i) Draw a graph to show the variation of the number of scattered particles detected (N) in Geiger-Marsden experiment as a function of scattering angle (θ).

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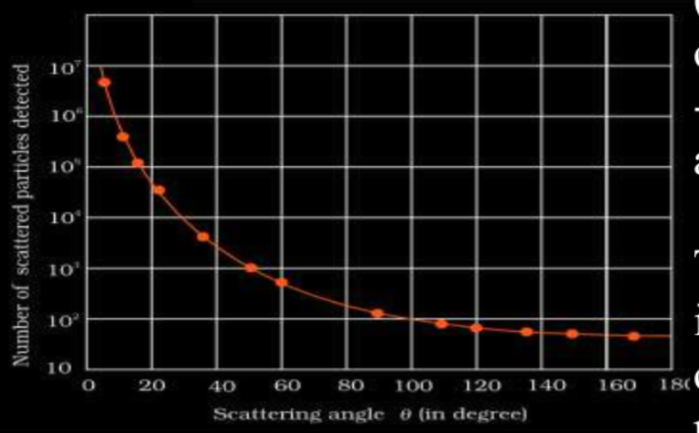
CBSE 2023 *3*



(ii)- The entire positive charge and most of the mass of the atom are concentrated in a small space. ½

-Many of the α-particles pass through the foil. It means that they do not suffer any collisions. ½

To deflect the α-particle backwards, a large repulsive force is required, which is provided only if the greater part of the mass of the atom & its positive charge were concentrated tightly at its centre. This lead to the discovery of the nucleus in the atom. 1

4. Hydrogen atom initially in the ground state, absorbs a photon which excites it to n = 5 level. The wavelength of the photon is :

(a) 975 nm (b) 740 nm

(c) 523 nm (d) 95 nm

CBSE 2023

4. Hydrogen atom initially in the ground state, absorbs a photon which excites it to n = 5 level. The wavelength of the photon is :

(a) 975 nm (b) 740 nm

(c) 523 nm (d) 95 nm

CBSE 2023

(d) 95 nm

22. The ground state energy of hydrogen atom is −13·6 eV in Bohr model. An electron makes a transition from an energy level of −1·51 eV to that of −3·4 eV. Calculate the wavelength of the spectral line emitted. CBSE 2023 *2*

22. The ground state energy of hydrogen atom is −13·6 eV in Bohr model. An electron makes a transition from an energy level of −1·51 eV to that of −3·4 eV. Calculate the wavelength of the spectral line emitted. CBSE 2023 *2*

$\Delta E = E_2 - E_1$ $= (-1.51-(-3.4))\text{eV}$ $= 1.89\text{eV}$	½
$=1.89 \times 1.6 \times 10^{-19}\,\text{J}$	½
$\Delta E = \frac{hc}{\lambda}$	½
$\lambda = \frac{hc}{\Delta E} = \frac{6.63\times10^{-34}\times3\times10^{8}}{1.89\times1.6\times10^{-19}} m$ $= 6.58\times10^{-7} m$ $= 658\text{nm}$	½

27. (a) Draw the energy level diagram for hydrogen atom. Mark the transitions corresponding to the series lying in the ultraviolet region, visible region and infrared region. CBSE 2023 *3*

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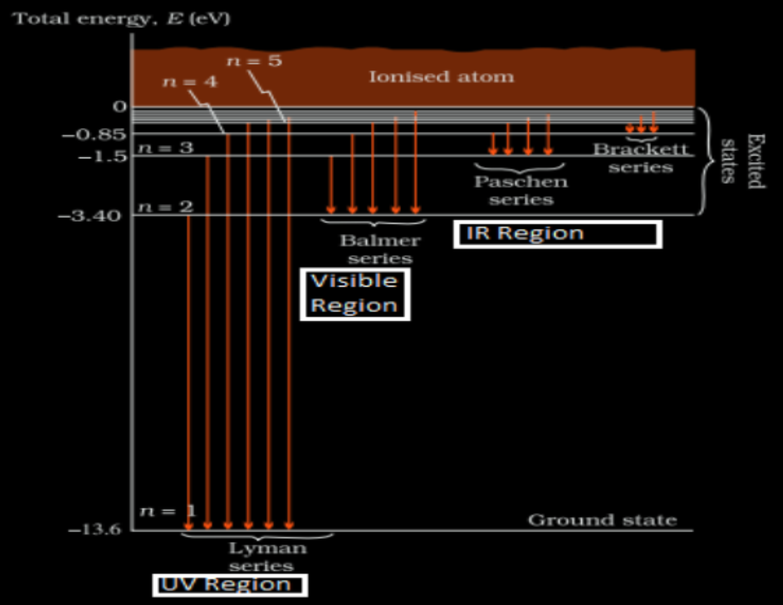


1 ½

½

½

½

Note: Award 1 ½ mark for energy level diagram if a student does not show the transitions.

10. The radius of the n^{th} orbit in Bohr model of hydrogen atom is proportional to :

(a) $\frac{1}{n^2}$ (b) $\frac{1}{n}$

(c) n^2 (d) n

CBSE 2023

10. The radius of the n^{th} orbit in Bohr model of hydrogen atom is proportional to :

(a) $\frac{1}{n^2}$ (b) $\frac{1}{n}$

(c) n^2 (d) n

CBSE 2023

(c) n^2

23. The ground state energy of hydrogen atom is $-13{\cdot}6$ eV. What is the potential energy and kinetic energy of an electron in the third excited state ?

CBSE 2023

2

23. The ground state energy of hydrogen atom is $-13{\cdot}6$ eV. What is the potential energy and kinetic energy of an electron in the third excited state ?

CBSE 2023 *2*

$E_n = \frac{-13.6}{n^2} eV$ = Total energy	
For third excited state n=4	
$E_4 = \frac{-13.6}{4^2} = \frac{-13.6}{16}$ = -0.85 eV	½
Potential Energy = 2 × Total Energy = 2 × E_4	
= 2 ×(-0.85) eV	
= - 1.70 eV	½
Kinetic energy = - (Total Energy) = $-E_4$	½
= 0.85 eV	½

20. (a) What is meant by ionisation energy ? Write its value for hydrogen atom ?

CBSE 2023 **2**

20. (a) What is meant by ionisation energy ? Write its value for hydrogen atom ? CBSE 2023 **2**

Ionization energy is the minimum energy required to remove an electron from an isolated atom of an element.	1
Alternatively	
It is the energy required to excite an electron from energy level n = 1 to n = ∞ from an isolated atom of an element.	
The ionization energy for hydrogen atom is 13.6 eV.	1

2. Specify the transition of electron in the wavelength of the line in the Bohr model of hydrogen atom which gives rise to the spectral line of highest wavelength. **1**

(a) $n = 3$ to $n = 1$ (b) $n = 3$ to $n = 2$

(c) $n = 4$ to $n = 1$ (d) $n = 4$ to $n = 2$

CBSE 2023

2. Specify the transition of electron in the wavelength of the line in the Bohr model of hydrogen atom which gives rise to the spectral line of highest wavelength. 1

(a) n = 3 to n = 1 (b) n = 3 to n = 2

(c) n = 4 to n = 1 (d) n = 4 to n = 2

CBSE 2023

(b) n=3 to n=2

11. A hydrogen atom makes a transition from n = 5 to n = 1 orbit. The wavelength of photon emitted is λ. The wavelength of photon emitted when it makes a transition from n = 5 to n = 2 orbit is **1**

(a) $\frac{8}{7}\lambda$

(b) $\frac{16}{7}\lambda$

(c) $\frac{24}{7}\lambda$

(d) $\frac{32}{7}\lambda$

CBSE 2023

11. A hydrogen atom makes a transition from n = 5 to n = 1 orbit. The wavelength of photon emitted is λ. The wavelength of photon emitted when it makes a transition from n = 5 to n = 2 orbit is **1**

(a) $\frac{8}{7}\lambda$

(b) $\frac{16}{7}\lambda$

(c) $\frac{24}{7}\lambda$

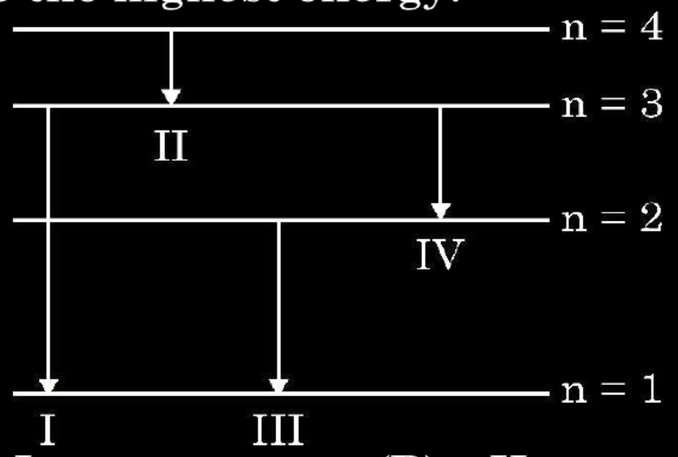
(d) $\frac{32}{7}\lambda$

CBSE 2023

(d)

$\frac{32}{7}\lambda$

10. The diagram shows four energy level of an electron in Bohr model of hydrogen atom. Identify the transition in which the emitted photon will have the highest energy. CBSE 2023 **1**



(A) I (B) II (C) III (D) IV

15. The radius of the n^{th} orbit in Bohr model of hydrogen atom is proportional to **1**

(A) n^2

(B) $\frac{1}{n^2}$

(C) n

(D) $\frac{1}{n}$

CBSE 2023

15. The radius of the n^{th} orbit in Bohr model of hydrogen atom is proportional to **1**

(A) n^2

(B) $\frac{1}{n^2}$

(C) n

(D) $\frac{1}{n}$

CBSE 2023

(A) n^2

30. (a) (i) Differentiate between 'distance of closest approach' and 'impact parameter'.

(ii) Determine the distance of closest approach when an alpha particle of kinetic energy 3.95 MeV approaches a nucleus of $Z = 79$, stops and reverses its directions. CBSE 2023 **3**

30. (a) (i) Differentiate between 'distance of closest approach' and 'impact parameter'.

(ii) Determine the distance of closest approach when an alpha particle of kinetic energy 3.95 MeV approaches a nucleus of Z = 79, stops and reverses its directions. CBSE 2023 **3**

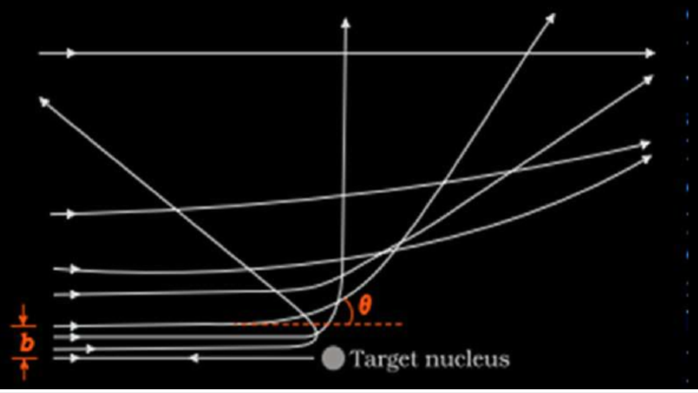
(i) **Distance of closest approach**: It is the minimum distance of α-particle from the centre of nucleus at which its total kinetic energy gets converted into electrostatic potential energy. **1**

Alternatively: Distance of closest approach

$$r_o = \frac{1}{4\pi \in_o}\frac{2Ze^2}{E_K}$$

Impact parameter: Perpendicular distance of the initial velocity vector of the α- particle from the centre of the nucleus. **1**

Alternatively:



(ii) $$r_o = \frac{1}{4\pi \in_o}\frac{2Ze^2}{E_K}$$ ½

$$r_o = \frac{9\times10^9\times2\times79\times(1.6\times10^{-19})^2}{3.95\times1.6\times10^{-13}}$$

$$= 57.6\times10^{-15}\,m$$ ½

(b) (i) State three postulates of Bohr's theory of hydrogen atom.

(ii) Find the angular momentum of an electron revolving in the second orbit in Bohr's hydrogen atom. CBSE 2023 **3**

(b) (i) State three postulates of Bohr's theory of hydrogen atom.

(ii) Find the angular momentum of an electron revolving in the second orbit in Bohr's hydrogen atom. CBSE 2023 **3**

(i) (1) An electron in an atom revolves in certain stable orbit without the emission of radiant energy.	½
(2) The electron revolves around the nucleus only in those orbits for which the angular momentum is integral multiple of $\frac{h}{2\pi}$ where h is the Planck's constant.	½
(3) When an electron makes a transition from one of its specified non-radiating orbits to another of lower energy orbit a photon is emitted having energy equal to the energy difference between the initial and final states.	½
(ii) Angular momentum $= \frac{nh}{2\pi}$	½
For n=2 Angular momentum $= \frac{2h}{2\pi}$	½
$= \frac{6.63\times10^{-34}}{\pi}$ $= 2.1\times10^{-34}\ kg\ m^2 s^{-1}$	½

4. (a) Define the term 'distance of closest approach'.

(b) An alpha particle approaches a nucleus with Z = 79. If its distance of closest approach is 28·8 fm, calculate the kinetic energy of the particle. CBSE 2022C *3*

(a)

Centre to centre distance between the α- particle and the nuclear when the kinetic energy of α- particle is completely converted to the potential energy of the system consisting α- particle and the nucleus. — 1

Alternatively

Centre to centre distance between the α- particle and the target nucleus when α- particle stops momentarily before rebounding.

Alternatively

When an α- particle approaches a target nucleus in a head-on position, the minimum distance of the α- particle from the centre of the target nucleus just before it rebounds is called distance of closest approach.

(b) $r_0 = \dfrac{2\,Z\,e^2}{4\,\pi\epsilon_0 E_k}$ — ½

$\therefore \quad E_k = \dfrac{2\,Z\,e^2}{4\,\pi\epsilon_0 r_0}$ — ½

$= \dfrac{9\times10^9\times2\times79\times(1.6\times10^{-19})^2}{28.8\times10^{-15}}$ — ½

$= 126.4\times10^{-14}\ \text{J}$

$= 7.9\ \text{MeV}$ — ½

1. (a) In Geiger-Marsden experiment, the distance of closest approach is considerably larger than the sum of the radii of the gold nucleus and the α-particle. Explain.

(b) The total energy of hydrogen atom in a state is – 3·4 eV. What are the kinetic and potential energies of the electron in this state ? *1+1*

CBSE 2022 C

1. (a) In Geiger-Marsden experiment, the distance of closest approach is considerably larger than the sum of the radii of the gold nucleus and the α-particle. Explain.

(b) The total energy of hydrogen atom in a state is – 3·4 eV. What are the kinetic and potential energies of the electron in this state ? *1+1*

CBSE 2022 C

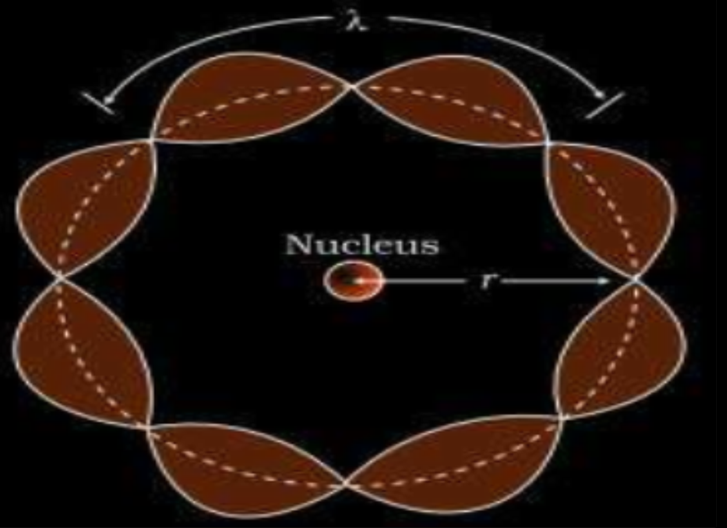
a) Alpha particle reverses its direction of motion , due to strong repulsive force, exerted by positively charged nucleus without even actually touching the gold nucleus.	1
b)E_n= -3.4eV (given) E_k= -E_n = 3.4eV U = 2E_n = - 6.8 eV	½ ½

9. (a) How did de Broglie hypothesis provide an explanation for Bohr's second postulate for quantisation of orbital angular momentum of the orbiting electron in hydrogen atom ? Discuss. CBSE 2022 C *2*

(b) Identify the transition of electron in Bohr model which gives rise to (i) the maximum, and (ii) the minimum wavelength in Balmer series of hydrogen spectrum. *1*

9. (a) How did de Broglie hypothesis provide an explanation for Bohr's second postulate for quantisation of orbital angular momentum of the orbiting electron in hydrogen atom ? Discuss. CBSE 2022 C *2*

(b) Identify the transition of electron in Bohr model which gives rise to (i) the maximum, and (ii) the minimum wavelength in Balmer series of hydrogen spectrum. *1*



For an electron moving in n^{th} circular orbit of radius r_n, the total distance is circumference of the orbit.

Thus $2\pi r_n = n\lambda$ (1) where n = 1, 2, 3 -------- ½

λ is de-broglie wavelength associated with the electron in the n^{th} orbit.

Now $\lambda = \frac{h}{p} = \frac{h}{mv_n}$ (2) ½

From equation (1) and (2)

$2\pi r_n = \frac{nh}{mv_n}$ ½

$mv_n r_n = \frac{nh}{2\pi}$ ½

Which is quantum condition proposed by Bohr for the angular momentum of the electron.

b) For Balmer series of hydrogen spectrum

$$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{n^2}\right)$$

(i) For maximum wavelength : transition of electron is from n = 3 to n=2 ½

(ii) For minimum wavelength: transition of electron is from n = ∞ to n=2 ½

3. (a) What results do you expect if α-particle scattering experiment is repeated using a thin sheet of hydrogen in place of a gold foil ? Explain. (Hydrogen is a solid at temperature below 14K) 2

CBSE 2022

3. (a) What results do you expect if α-particle scattering experiment is repeated using a thin sheet of hydrogen in place of a gold foil ? Explain. (Hydrogen is a solid at temperature below 14K) 2

CBSE 2022

There would be no large angle scattering / Size of nucleus can't be determined.	1
As hydrogen atom which is a target nucleus, has only one proton whereas approaching α particle is more massive than the target nucleus.	1
Alternatively: Repulsive force between target nucleus (Hydrogen) and α–particles will be very less.	
(**Note:** Give full credit for other correct explanations.)	

5. An electron in a hydrogen atom makes transitions from orbits of higher energies to orbits of lower energies. 3

(i) When will such transitions result in (a) Lyman (b) Balmer series ?

(ii) Find the ratio of the longest wavelength in Lyman series to the shortest wavelength in Balmer series.

CBSE 2022

i) In Lyman series transition takes place from

$n_i = 2, 3 - - - - - \infty$ to $n_f = 1$ ½

In Balmer series transition takes place from ½

$n_i = 3, 4 - - - - - \infty$ to $n_f = 2$

ii) Longest wavelength in Lyman series (λ_1)

$\frac{1}{\lambda_1} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$ ½

$n_f = 1 \qquad n_i = 2$

$\frac{1}{\lambda_1} = R\left(\frac{1}{1^2} - \frac{1}{2^2}\right)$

$\frac{1}{\lambda_1} = \frac{3R}{4}$ ----------(i) ½

Shortest wavelength in Balmer series (λ_2)

$\frac{1}{\lambda_2} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$

$n_f = 2 \qquad n_i = \infty$

$\frac{1}{\lambda_2} = R\left(\frac{1}{2^2} - \frac{1}{\infty}\right) = \frac{R}{4}$ ½

$\frac{\lambda_1}{\lambda_2} = \frac{1}{3}$ ½

9. (a) Use Bohr's postulate to prove that the radius of n^{th} orbit in a hydrogen atom is proportional to n^2. 3

(b) How will the energy of a hydrogen atom change if n increases from 1 to ∞ ?

CBSE 2022

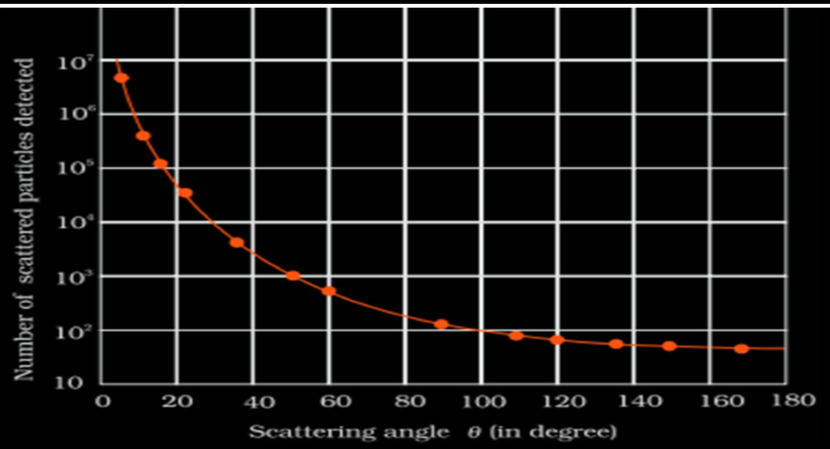
(a) centripetal force = electrostatic force of attraction	
$\frac{mv^2}{r} = \frac{ke^2}{r^2}$	½
$r = \frac{ke^2}{mv^2}$ --------- (i)	
From Bohr's II postulate of quantization	
$L = mvr = \frac{nh}{2\pi}$ ---------- (ii)	½
Substitute v from eq (ii) into eq (i)	
$r = \frac{ke^2}{m \cdot n^2h^2} \times 4\pi^2m^2r^2$	½
$r = \frac{n^2h^2}{4\pi^2mKe^2}$	
$r \alpha n^2$	½
(b) $E_n = \frac{-13.6}{n^2}$ eV	½
For n =1, E_1= -13.6eV, for n = ∞, E_∞=0	
As n increases from n = 1 to ∞,energy increases (Note: Give full credit of part (b) if a student does not write the formula)	½

4. Draw a graph showing the variation of number of particles scattered (N) with the scattering angle θ in Geiger-Marsden experiment. Why only a small fraction of the particles are scattered at $\theta > 90^{o}$? Mention two limitations of Rutherford nuclear model of an atom. 3

CBSE 2022

4. Draw a graph showing the variation of number of particles scattered (N) with the scattering angle θ in Geiger-Marsden experiment. Why only a small fraction of the particles are scattered at $\theta > 90^{o}$? Mention two limitations of Rutherford nuclear model of an atom. 3

CBSE 2022



1

Due to the large empty space available in an atom, α particles heading towards the nucleus are less, therefore, small fraction of particles are scattered at $\theta > 90^{0}$ 1

Alternatively:

For angle $\theta > 90^{0}$, the impact parameter is very small.

Limitations

- Could not explain the stability of the atom ½
- Could not explain the existence of line spectra ½

11. (a) The energy of hydrogen atom in an orbit is -1.51 eV. What are kinetic and potential energies of the electron in this orbit ?

(b) The electron in a hydrogen atom is typically found at a distance of about 5.3×10^{-11} m from the nucleus which has a diameter of about 1.0×10^{-15} m. Assuming the hydrogen atom to be a sphere of radius 5.3×10^{-11} m, what fraction of its volume is occupied by the nucleus ? CBSE 2022 **3**

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E= -1.51eV	
$K = -E$	½
=1.51 eV	½
$U = 2E$	½
= -3.02eV	½
Fraction of volume occupied by the nucleus	
$= \frac{\frac{4}{3}\pi r^3 nucleus}{\frac{4}{3}\pi r^3 atom}$	½
$= \left(\frac{r_{nucleus}}{r_{atom}}\right)^3$	
$= \left(\frac{0.5 x 10^{-15}}{5.3 x 10^{-11}}\right)^3$	
$= 8.3 \times 10^{-16}$	½

7. State Bohr's postulate to explain stable orbits in a hydrogen atom. Prove that the speed with which the electron revolves in n^{th} orbit is proportional to (1/n). CBSE 2022 **3**

7. State Bohr's postulate to explain stable orbits in a hydrogen atom. Prove that the speed with which the electron revolves in n^{th} orbit is proportional to (1/n). CBSE 2022 **3**

(i) An electron can revolve around the nucleus in an orbit in which its angular momentum is an integral multiple of $\frac{h}{2\pi}$.	1
(ii) Proof	
$\frac{mv^2}{r} = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2}$	½
$mvr = \frac{nh}{2\pi}$	½
Eliminating r we get	
$v = \frac{e^2}{2\varepsilon_0 h}\cdot\frac{1}{n}$	½
$\therefore v \propto \frac{1}{n}$	½

9. A narrow beam of protons, each having 4.1 MeV energy is approaching a sheet of lead (Z = 82). Calculate :

(i) the speed of a proton in the beam, and

(ii) the distance of its closest approach

CBSE 2022 **3**

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(i) the speed of a proton in the beam, and

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CBSE 2022 **3**

(i) $\frac{1}{2}mv^2 = 4{\cdot}1\times1{\cdot}6\times10^{-13}\,\text{J}$ ½

$$v = \sqrt{\frac{2\times4{\cdot}1\times1.6\times10^{-13}}{1{\cdot}673\times10^{-27}}}$$ ½

$= 2{\cdot}8\times10^{7}$ m/s ½

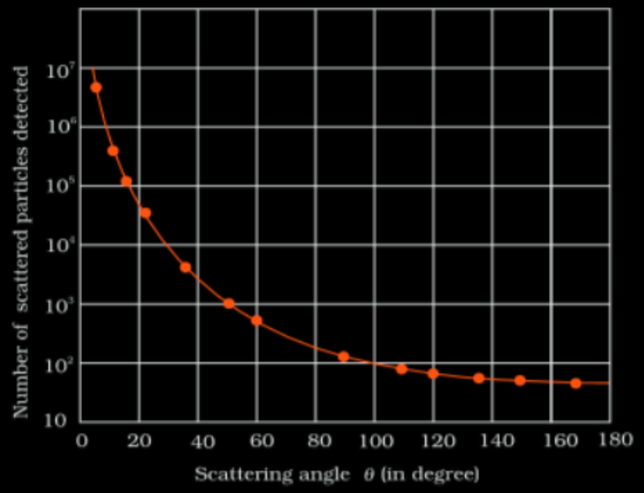
(ii) $d = \frac{Ze^2}{4\pi\varepsilon_0\times E_k}$ ½

$= \frac{9\times10^{9}\times82\times1{\cdot}6\times10^{-19}\times1{\cdot}6\times10^{-19}}{4{\cdot}1\times1{\cdot}6\times10^{-13}}$ ½

$= 2.88\ \text{X}\ 10^{-14}$ m ½

1. (a) Draw the graph showing the variation of the number (N) of scattered alpha particles with scattering angle (θ) in Geiger – Marsden experiment. Infer two conclusions from the graph. CBSE 2022 **2**

1. (a) Draw the graph showing the variation of the number (N) of scattered alpha particles with scattering angle (θ) in Geiger – Marsden experiment. Infer two conclusions from the graph. CBSE 2022 **2**



1

(Give full credit if axis are marked and values are not given)

Conclusions

- Most of the alpha particles pass undeviated through the gold foil.
- A few alpha particles , get deflected through 90° or more.
- Only about 0.14% of the incident alpha particles are reflected by large angle.
- A very few alpha particles retrace their path.

½+½

Any other two conclusions

5. Define the term – Distance of closest approach. How will it be affected, for an α - particle, if kinetic energy of the particle is doubled ? CBSE 2022 **3**

5. Define the term – Distance of closest approach. How will it be affected, for an α - particle, if kinetic energy of the particle is doubled ? CBSE 2022 **3**

The minimum distance up to which an alpha particle travel along the central line of the nucleus before it rebounds is called distance of closest approach.	1 ½
Alternatively	
An alpha particle travelling directly towards the centre of a nucleus slows down as it approaches the nucleus due to repulsive force. At a distance r_0 from the nucleus, the α- particle stops and its total kinetic energy converts into electrostatic potential energy. This distance r_0 is called distance of closest approach.	
$r_0 = \frac{2Ze^2}{4\pi \in_0 K.E.}$	½
i.e., $r_0 \propto \frac{1}{K.E.}$	½
as K is doubled , r_0 is halved.	½
(Award full 1½ marks if a student writes r_0 is halved without writing formula)	

9. (a) Sketch the energy level diagram for hydrogen atom. **3**

(b) Find the ratio of the longest and the shortest wavelength in Lyman series in hydrogen atom.

CBSE 2022

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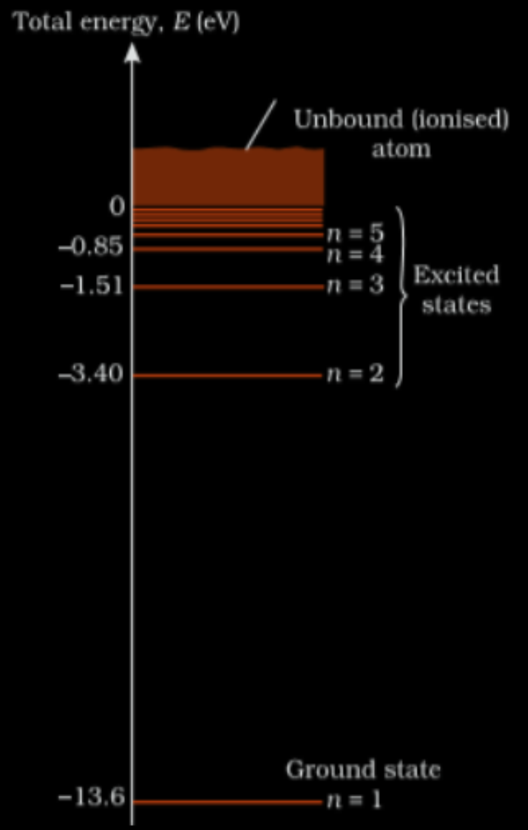
CBSE 2022

i) Hydrogen atom energy level diagram



1

ii) $\frac{1}{\lambda} = R\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$ ½

For longest wavelength in Lyman series

$n_1 = 1,\ n_2 = 2$

$$\frac{1}{\lambda_1} = R\left[\frac{1}{1^2} - \frac{1}{2^2}\right] = R\left[1 - \frac{1}{4}\right] = \frac{3}{4}R$$ ½

$$\therefore \lambda_1 = \frac{4}{3R}$$

For shortest wavelength in Lyman Series

$n_1 = 1,\ n_2 = \infty$ ½

$$\frac{1}{\lambda_2} = R\left[\frac{1}{1^2} - \frac{1}{\infty^2}\right] = R[1-0] = R$$

$$\therefore \lambda_2 = \frac{1}{R}$$

$$\frac{\lambda_1}{\lambda_2} = \frac{\frac{4}{3R}}{\frac{1}{R}} = \frac{4}{3}$$ ½

2. (a) Define the terms 'distance of closest approach' and 'impact parameter'. CBSE 2022 *2*

2. (a) Define the terms 'distance of closest approach' and 'impact parameter'. CBSE 2022 *2*

Definition of: Distance of closest approach: The minimum distance of the approaching particle from the nucleus. Impact parameter : The perpendicular distance of the approaching particle from the centre of a nucleus when the particle is far away from the nucleus.	1 1

4. A hydrogen atom has only one electron yet its spectrum has many lines. Explain. Name the spectral series which lies in the visible region of electromagnetic spectrum. Mention the conditions under which the spectral line in this series will have (a) maximum, and (b) minimum wavelengths. CBSE 2022 *3*

4. A hydrogen atom has only one electron yet its spectrum has many lines. Explain. Name the spectral series which lies in the visible region of electromagnetic spectrum. Mention the conditions under which the spectral line in this series will have (a) maximum, and (b) minimum wavelengths. CBSE 2022 *3*

• The electron may remain present in different excited states of hydrogen atom and then it jumps back to the lower energy states or the ground state.	1
• Balmer series	1
• Maximum wavelength of the line in Balmer series corresponds to the transition from n=3 to n=2	½
• Minimum wavelength of the line corresponds to the transition from n=∞ to n=2	½

6. (a) State Bohr's second postulate for hydrogen atom.

(b) The ground state energy of a hydrogen atom is –13·6 eV. Find its kinetic energy and potential energy in the ground and the second excited states. CBSE 2022 *3*

6. (a) State Bohr's second postulate for hydrogen atom.

(b) The ground state energy of a hydrogen atom is –13·6 eV. Find its kinetic energy and potential energy in the ground and the second excited states. CBSE 2022 *3*

Statement The electron can revolve around the nucleus in an orbit in which its angular momentum is integral multiple of $\frac{h}{2\pi}$	1
In the ground state	
$KE = -E = +13{\cdot}6\ eV$	½
$PE = -2\ KE = -27{\cdot}2\ eV$	½
In the second excited state, $n = 3$	
$E_2 = -\frac{13{\cdot}6}{n^2} = -\frac{13{\cdot}6}{3^2} = -1{\cdot}51\ eV$	½
$KE = -E_2 = +1{\cdot}51\ eV$	
$PE = -2\ KE = -2\times1.51 = -3.02\ eV$	½

1. (a) Name the spectral series for a hydrogen atom which lies in the visible region. Find the ratio of the maximum to the minimum wavelengths of this series. CBSE 2022 *2*

1. (a) Name the spectral series for a hydrogen atom which lies in the visible region. Find the ratio of the maximum to the minimum wavelengths of this series. CBSE 2022 *2*

Balmer Series	½
$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$	½
$\frac{1}{\lambda_{max}} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$ $(n_f = 2, \quad n_i = 3)$	½
$\lambda_{max} = \frac{36}{5R}$	
$\frac{1}{\lambda_{min}} = R\left(\frac{1}{2^2} - \frac{1}{\infty}\right)$ $(n_f = 2, \quad n_i = \infty)$	
$\lambda_{min} = \frac{4}{R}$	
$\frac{\lambda_{max}}{\lambda_{min}} = \frac{9}{5}$	½

9. (a) In Geiger-Marsden experiment, calculate the distance of closest approach for an alpha particle with energy $2{\cdot}56 \times 10^{-12}$ J. Consider that the particle approaches gold nucleus ($Z = 79$) in head-on position.

(b) If the above experiment is repeated with a proton of the same energy, then what will be the value of the distance of closest approach ?

CBSE 2022 *3*

a) For α particles, distance of closest approach

$$r_\alpha = \frac{1}{4\pi\varepsilon_0}\frac{2Ze^2}{E_k} \qquad \text{½}$$

$$r_\alpha = \frac{9\times10^9\times2\times79\times(1.6\times10^{-19})^2}{2.56\times10^{-12}} \qquad \text{½}$$

$$= 14{\cdot}22\times10^{-15}\ \text{m}$$

$$= 14.22\ \text{fm} \qquad \text{½}$$

b) For proton, distance of closest approach

$$r_p = \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{E_k} \qquad \text{½}$$

$$r_p = \frac{r_\alpha}{2} \qquad \text{½}$$

$$= 7{\cdot}11\times10^{-15}\ \text{m} \qquad \text{½}$$

3. (a) (i) Define the terms : 'impact parameter' and 'distance of closest approach' for an α-particle in Geiger-Marsden scattering experiment.

(ii) What will be the value of the impact parameter for scattering angle (I) $\theta = 0°$ and (II) $\theta = 180°$? CBSE 2022 *2*

3. (a) (i) Define the terms : 'impact parameter' and 'distance of closest approach' for an α-particle in Geiger-Marsden scattering experiment.

(ii) What will be the value of the impact parameter for scattering angle (I) $\theta = 0°$ and (II) $\theta = 180°$? CBSE 2022 *2*

i. **Impact Parameter** : It is the perpendicular distance of the initial velocity vector of the approaching α-particle from the centre of the nucleus.	½
Distance of closest approach : It is the minimum distance of the approaching α-particle and the target gold nucleus	½
$d = \frac{2Ze^2}{4\pi\varepsilon_0 K}$; Where K is the kinetic energy	
Alternatively : Distance of closest approach is the distance of the alpha particle from the centre of gold nucleus where its whole kinetic energy is converted into potential energy	
ii. $\theta = 0°$; b = maximum / almost of atomic size	½
$\theta = 180°$; b = minimum = zero	½
(Note: Allot ½ Mark for only formula.)	

11. An electron is in the third excited state in a hydrogen atom. It undergoes transitions to the lower energy states.

(a) What is the maximum number of spectral lines that can be emitted ?

(b) Calculate the minimum wavelength of the spectral lines emitted. *3*

CBSE 2022

11. An electron is in the third excited state in a hydrogen atom. It undergoes transitions to the lower energy states.

(a) What is the maximum number of spectral lines that can be emitted ?

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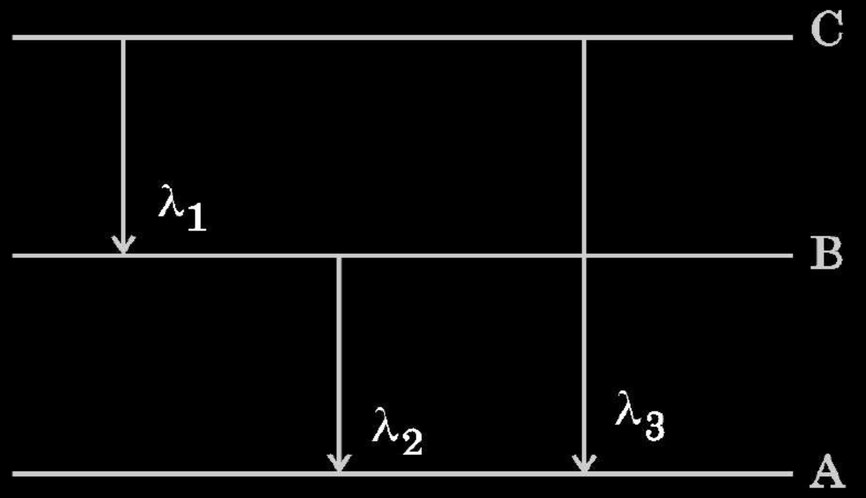
CBSE 2022

(a) 6	1
(b)	
$hv = \frac{hc}{\lambda} = E_4 - E_1$	½
$= -0.85 + 13.6$	½
$= +12.75 ev$	
$\frac{hc}{\lambda} = 12.75 \times 1.6 \times 10^{-19}$	
$\lambda = \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{12.75 \times 1.6 \times 10^{-19}}$	½
$\lambda = 975 nm = 9.75 \times 10^{-11} m$	½

9. (a) State Bohr's quantization condition for defining stationary orbits.

(b) Use the energy level diagram shown below to obtain the relation between three wavelengths λ_1, λ_2 and λ_3 emitted due to the transition of electron from the energy states C and B. CBSE 2022 *3*



a) Condition for stationary orbits:- The electron revolves around the nucleus only in those orbits for which angular momentum is some integral multiple of $\frac{h}{2\pi}$. — 1

$$mvr = L = \frac{nh}{2\pi r}$$

(b) For transition $C \to A$ $\quad \frac{hc}{\lambda_3} = E_C - E_A$ — ½

For transition $C \to B$ $\quad \frac{hc}{\lambda_1} = E_C - E_B$ — ½

For transition $B \to A$ $\quad \frac{hc}{\lambda_2} = E_B - E_A$ — ½

$$\Rightarrow E_C - E_A = E_C - E_B + E_B - E_A$$

$$\frac{1}{\lambda_3} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$$ — ½

8. The energy of a hydrogen atom in the first excited state is – 3·4 eV. Find :

CBSE 2022

(a) the radius of this orbit. (Take Bohr radius = 0·53 Å)

(b) the angular momentum of the electron in the orbit.

(c) the kinetic and potential energy of the electron in the orbit. *3*

8. The energy of a hydrogen atom in the first excited state is – 3·4 eV. Find :

CBSE 2022

(a) the radius of this orbit. (Take Bohr radius = 0·53 Å)

(b) the angular momentum of the electron in the orbit.

(c) the kinetic and potential energy of the electron in the orbit. *3*

a) $r = n^2 r_0 = (2)^2 \times 0.53 = 2.12$ Å	1
b) $L = \frac{nh}{2\pi} = \frac{2 \times 6.63 \times 10^{-34}}{2 \times 3.14} = 2 \times 10^{-34}$ kg m/s	1
c) K.E = + 3.4 eV & P.E = - 6.8eV	1

26. Use Bohr's model of hydrogen atom to obtain the relationship between the angular momentum and the magnetic moment of the revolving electron. **2**

CBSE 2020

26. Use Bohr's model of hydrogen atom to obtain the relationship between the angular momentum and the magnetic moment of the revolving electron. **2**

CBSE 2020

According to Bohr's model	
$L = Angular\ momentum = mvr = \frac{nh}{2\pi}$	½
$\mu = Magnetic\ moment = current \times area\ of\ the\ orbit$	½
$\mu = \lvert e \rvert \times \nu \times \pi r^2 = \frac{\lvert e \rvert vr}{2}$	½
$\therefore \frac{L}{\mu} = \frac{mvr \times 2}{\lvert e \rvert vr} = \frac{2m}{\lvert e \rvert}$	
$\mu = \frac{\lvert e \rvert}{2m} L$	½

29. Calculate the de-Broglie wavelength associated with the electron revolving in the first excited state of hydrogen atom. The ground state energy of the hydrogen atom is –13.6 eV. CBSE 2020 **3**

29. Calculate the de-Broglie wavelength associated with the electron revolving in the first excited state of hydrogen atom. The ground state energy of the hydrogen atom is –13.6 eV. CBSE 2020 **3**

Energy of the electron in the first excited state

$$E_1 = -\frac{13.6}{2^2} eV = -3.4eV \quad \text{½}$$

$$= -3.4 \times 1.6 \times 10^{-19} J$$

$$= -5.44 \times 10^{-19} J \quad \text{½}$$

Associated kinetic energy =Negative of total energy ½

$K = 5.44 \times 10^{-19} J$ ½

de-Broglie wavelength, $\lambda = h/p$

$$\lambda = \frac{h}{\sqrt{2mK}} \quad \text{½}$$

$$\lambda = \frac{6.63 \times 10^{-34}}{(2 \times 9.1 \times 10^{-31} \times 5.44 \times 10^{-19})^{1/2}} \text{ m}$$

$$\lambda = \frac{6.63 \times 10^{-34}}{(99.008)^{1/2} \times 10^{-25}} \text{ m}$$

$\approx 0.663 \times 10^{-9}$ m $= 0.663$nm $= 6.63A^{\circ}$ ½

9. Paschen series of atomic spectrum of hydrogen gas lies in : CBSE 2020

(a) Infrared region
(b) Ultraviolet region
(c) Visible region
(d) Partly in ultraviolet and partly in visible region **1**

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(a) Infrared region
(b) Ultraviolet region
(c) Visible region
(d) Partly in ultraviolet and partly in visible region **1**

(a) Infra red region

10. In the α–particle scattering experiment, the shape of the trajectory of the scattered α–particles depend upon : CBSE 2020

(a) only on impact parameter.
(b) only on the source of α–particles.
(c) both impact parameter and source of α–particles.
(d) impact parameter and the screen material of the detector. **1**

10. In the α–particle scattering experiment, the shape of the trajectory of the scattered α–particles depend upon :

CBSE 2020

(a) only on impact parameter.
(b) only on the source of α–particles.
(c) both impact parameter and source of α–particles.
(d) impact parameter and the screen material of the detector. **1**

(a) Only on impact parameter

(a) State the postulates of Bohr's model of hydrogen atom and derive the expression for Bohr radius.

(b) Find the ratio of the longest and the shortest wavelengths amongst the spectral lines of Balmer series in the spectrum of hydrogen atom. **5**

CBSE 2020

(a) Bohr's Postulates:-

1) An electron in an atom could revolve in certain stable orbits without the emission of radiant energy ½
2) The electron revolves around the nucleus only in those orbits for which the angular momentum is some integral multiple of h/2π where h is the Planck's constant ½
3) The frequency of the emitted photon when an electron makes a transition from higher orbit to lower energy orbit is given by

$h\nu = E_2 - E_1$

$L_n = mv_n r_n = \frac{nh}{2\pi}$ (i) ½

$\frac{mv_n^2}{r_n} = \frac{1}{4\pi\epsilon_0}\frac{e^2}{r_n^2}$ ½

$$v_n = \frac{e}{\sqrt{4\pi\epsilon_0 m r_n}}$$ ½

Combining wit h equation (i)

$$v_n = \frac{1}{n}\frac{e^2}{4\pi\epsilon_0}\frac{1}{(h/\pi)}$$

$$r_n = \left(\frac{n^2}{m}\right)\left(\frac{h}{2\pi}\right)^2 \left(\frac{4\pi\epsilon_0}{e^2}\right)$$ ½

(b) For shortest wave length

$\frac{1}{\lambda_S} = R(\frac{1}{2^2} - \frac{1}{\infty})$

$\frac{1}{\lambda_S} = \frac{R}{4}$ (i) ½

For longest wave length

$\frac{1}{\lambda_L} = R(\frac{1}{2^2} - \frac{1}{3^2})$

$= R(\frac{1}{4} - \frac{1}{9})$

$= R(\frac{5}{36})$ (ii) ½

Dividing equation (i) by equation (ii) we get

$$\frac{(1/\lambda_S)}{(1/\lambda_L)} = \frac{(R/4)}{(5R/36)}$$ ½

$\frac{\lambda_L}{\lambda_S} = \frac{9}{5}$ OR $\lambda_L : \lambda_S = 9:5$ ½

7. In Bohr's model of hydrogen atom, the total energy of the electron in n^{th} discrete orbit is proportional to *1*

(A) n

(B) $\frac{1}{n}$

(C) n^2

(D) $\frac{1}{n^2}$

CBSE 2020

7. In Bohr's model of hydrogen atom, the total energy of the electron in n^{th} discrete orbit is proportional to *1*

(A) n

(B) $\frac{1}{n}$

(C) n^2

(D) $\frac{1}{n^2}$

CBSE 2020

(D)

$\frac{1}{n^2}$

---

27. A hydrogen atom is in its third excited state.

(a) How many spectral lines can be emitted by it before coming to the ground state ? Show these transitions in the energy level diagram.

(b) In which of the above transitions will the spectral line of shortest wavelength be emitted ?

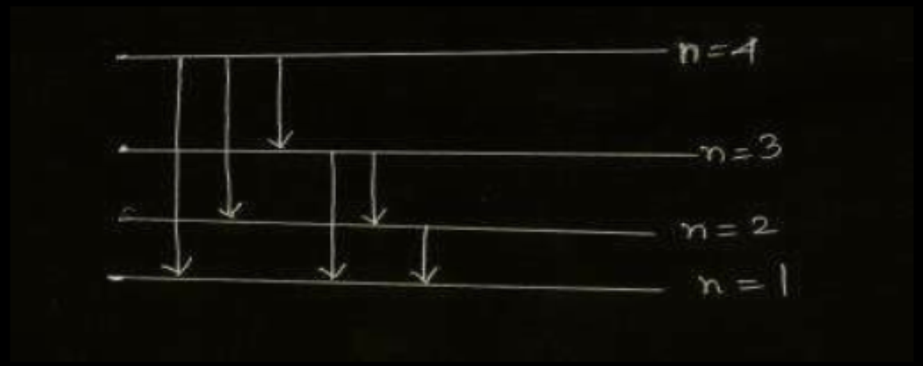
CBSE 2020 *2*

27. A hydrogen atom is in its third excited state.

(a) How many spectral lines can be emitted by it before coming to the ground state ? Show these transitions in the energy level diagram.

(b) In which of the above transitions will the spectral line of shortest wavelength be emitted ?

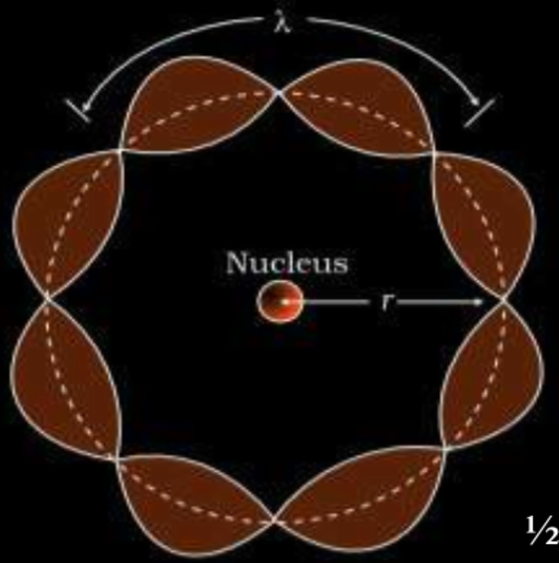
CBSE 2020 *2*

a) number of spectral lines =6 energy level diagram  b) n=4 to n=1	½ 1 ½

32. What do you mean by wave nature of an electron ? How was quantisation of angular momentum of the orbiting electron in Bohr's model of hydrogen atom explained by de Broglie hypothesis ? CBSE 2020 *3*

32. What do you mean by wave nature of an electron ? How was quantisation of angular momentum of the orbiting electron in Bohr's model of hydrogen atom explained by de Broglie hypothesis ? CBSE 2020 *3*

Moving electron can show wave characteristics. 1



½

From the diagram

$$2\pi r = n\lambda$$ ½

(Note: Award one mark here even if the student just writes this equation without drawing the diagram)

According to de Broglie

$$\lambda = \frac{h}{p}$$ ½

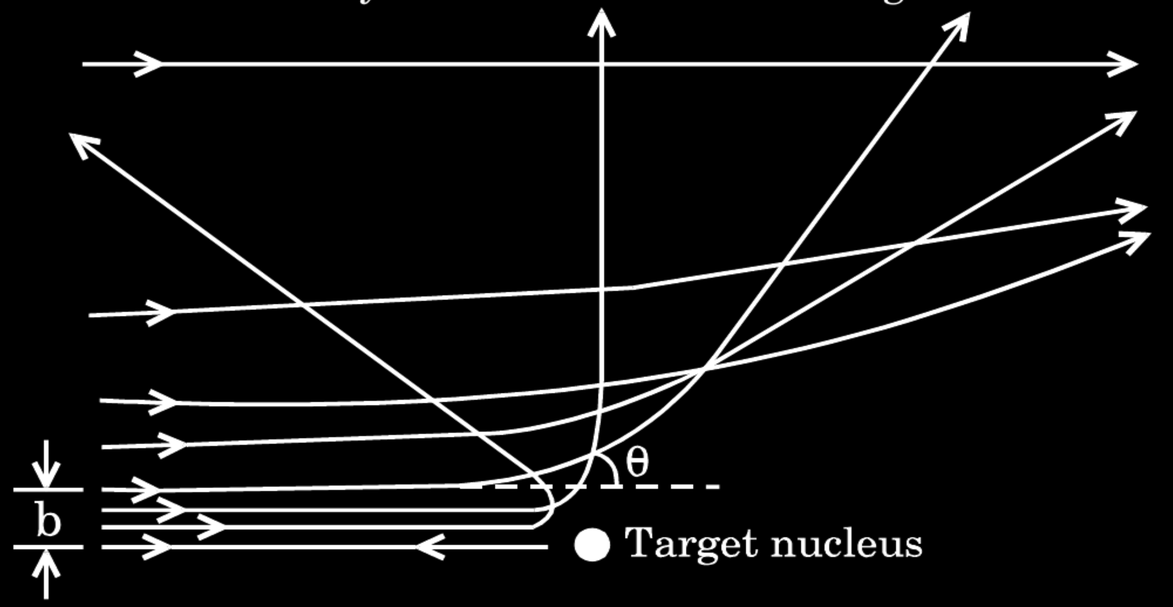
$$\therefore 2\pi r = n\lambda = \frac{nh}{p}$$

$$2\pi r = \frac{nh}{mv}$$

$$mvr = \frac{nh}{2\pi} \; where \; n = 1,2,3, \ldots ..$$

This explains the quantisation of angular momentum of the orbiting electron. ½

26. In Geiger-Marsden scattering experiment, the trajectory of α-particles in Coulomb's field of a heavy nucleus is shown in the figure.



(a) What do 'b' and 'θ' represent in the figure ? CBSE 2020

(b) What will be the value of 'b' for (i) $\theta = 0°$, and (ii) $\theta = 180°$? *2*

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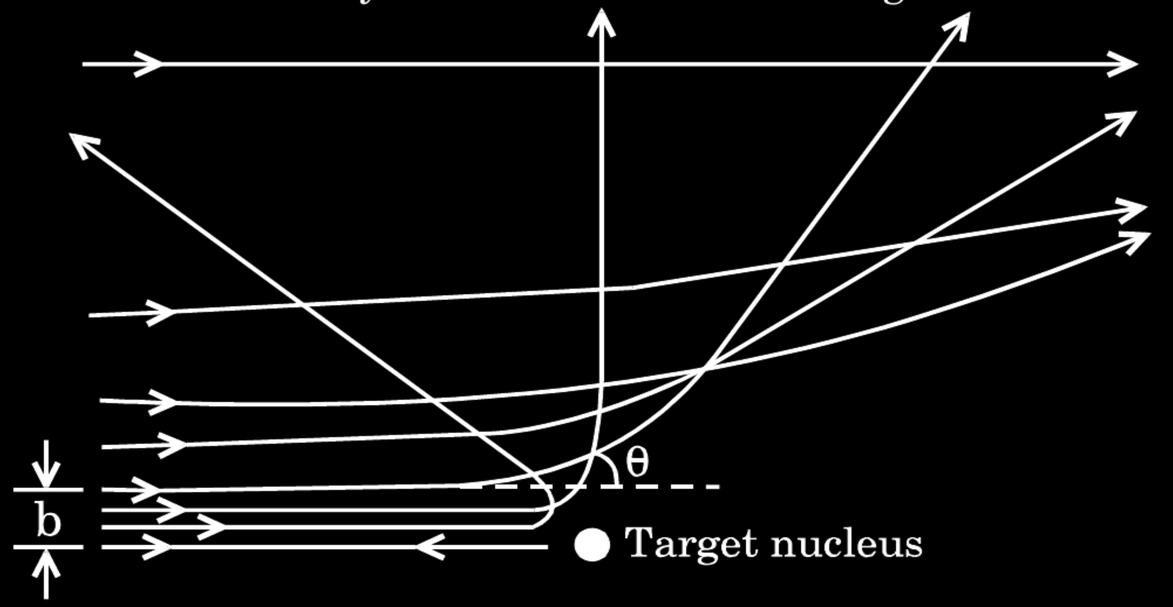

(a) What do 'b' and 'θ' represent in the figure ? CBSE 2020

(b) What will be the value of 'b' for (i) $\theta = 0°$, and (ii) $\theta = 180°$? *2*

(a) b represents impact parameter	½
θ represents scattering angle	½
(b) (i) for $\theta = 0^0$ impact parameter is large or infinite	½
(ii) for $\theta = 180^0$ impact parameter is zero.	½

9. Which of the following statements is ***not*** correct according to Rutherford model ? *1*

(A) Most of the space inside an atom is empty.

(B) The electrons revolve around the nucleus under the influence of coulomb force acting on them.

(C) Most part of the mass of the atom and its positive charge are concentrated at its centre.

(D) The stability of atom was established by the model.

CBSE 2020

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(D)

The stability of atom was established by the model

14. According to Bohr's atomic model, the circumference of the electron orbit is always an _____ multiple of de Broglie wavelength. CBSE 2020 *1*

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Integral
OR
Nucleons

26. Using Bohr's atomic model, derive the expression for the radius of n^{th} orbit of the revolving electron in a hydrogen atom. CBSE 2020 *2*

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$$F_c = F_E$$

$$\frac{m_e v_n^2}{r_n} = \frac{Kze^2}{r_n^2} \qquad ½$$

$$m_e v_n^2 r_n = Kze^2$$

By Bohr's second postulate

$$L = m_e v_n r_n = \frac{nh}{2\pi} \qquad ½$$

$$r_n = \frac{n^2 h^2}{4\pi^2 m_e k e^2 Z}$$

$$r_n = \frac{n^2 h^2}{4\pi^2 m_e k e^2} \ (\because Z = 1) \qquad 1$$

Obtain the first Bohr's radius and the ground state energy of a muonic hydrogen atom i.e. an atom where the electron is replaced by a negatively charged muon (μ^-) of mass about 207 m_e that orbits around a proton. (Given for hydrogen atom, radius of first orbit and ground state energy are $0{\cdot}53 \times 10^{-10}$ m and $-13{\cdot}6$ eV respectively)

CBSE 2019

In Bohr's Model of hydrogen atom the radius of n^{th} orbit is given by

$$r_n = \frac{n^2 h^2}{4\pi^2 e^2 \ m_e}$$

$$r_1 \ \alpha \ \frac{1}{m_e} \quad \text{(as n=1)}$$

Similarly

$$r_\mu \propto \frac{1}{m_u}$$

$$\frac{r_\mu}{r_e} = \frac{m_e}{m_\mu} = \frac{1}{207}$$

$\therefore r_\mu = 2.56 \times 10^{-13}$ m

Energy of electron in n^{th} orbit

$$E_n = \frac{-2\pi^2 me^4}{n^2 h^2}$$

$E_n \ \alpha \ m_e$ (as n=1)

$$\therefore \frac{E_\mu}{E_e} = \frac{m_\mu}{m_e} = 207$$

$$\therefore E_\mu = 207 \ E_e$$

$= - 207 \times 13.6$ eV

$= - 2.8$ keV

[Even if a student writes correct expressions for r_n and E_n or correct proportionality relation $r_n \ \alpha \ n^2$ and $E_n \ \alpha \ 1/n^2$ award full mark]

(a) Using Bohr's postulates, derive the expression for the orbital period of the electron moving in the n^{th} orbit of hydrogen atom.

(b) Write Rydberg's formula for wavelengths of the spectral lines of hydrogen atom spectrum. Mention to which series in the emission spectrum of hydrogen, H_α line belongs.

CBSE 2019

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CBSE 2019

a) Orbital Period of electron in hydrogen atom

$$T = \frac{2\pi r_n}{v_n} \quad \text{..... (eq}^n\text{1)}$$ ½

From Bohr's postulates

$$mv_n r_n = \frac{nh}{2\pi}$$

We have $r_n = \frac{n^2h^2\varepsilon_0}{\pi me^2}$ ½

and velocity of electron in n^{th} orbital

$$v_n = \frac{e^2}{2\varepsilon_0 nh}$$ ½

On substituting the values of r_n and v_n in equation (1) we have

$$T = \frac{4n^2h^2\varepsilon_0}{me^4}$$ ½

b) Rydberg's formula for wavelengths of the spectral lines

$$\frac{1}{\lambda} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$ ½

H_α belongs to **Balmer series**. ½

Obtain Bohr's quantisation condition for angular momentum of electron orbiting in n^{th} orbit in hydrogen atom on the basis of the wave picture of an electron using de Broglie hypothesis.

CBSE 2019

Obtain Bohr's quantisation condition for angular momentum of electron orbiting in n^{th} orbit in hydrogen atom on the basis of the wave picture of an electron using de Broglie hypothesis.

CBSE 2019

Let λ be the deBroglie wavelength associated with electron orbiting (with speed v) in the with n^{th} orbit (of radius r) in hydrogen atom.	
$\therefore \quad \lambda = \frac{h}{mv}$	½
Also $2\pi r = n\lambda$	½
$= \frac{nh}{mv}$	
$\therefore \quad L = mvr = \frac{nh}{2\pi}$	1

A hydrogen atom in the ground state is excited by an electron beam of 12·5 eV energy. Find out the maximum number of lines emitted by the atom from its excited state.

CBSE 2019

A hydrogen atom in the ground state is excited by an electron beam of 12·5 eV energy. Find out the maximum number of lines emitted by the atom from its excited state.

CBSE 2019

Energy in ground state, $E_1 = -13.6eV$	
Energy supplied $= 12.5eV$	
Energy in excited state, $-13.6 + 12.5 = -1.1eV$	½
But, $E_n = \frac{-13.6}{n^2} = -1.1$	½
$n = 3$	½
Maximum number of lines=3	½

Prove that the magnetic moment of the electron revolving around a nucleus in an orbit of radius r with orbital speed v is equal to evr/2. Hence using Bohr's postulate of quantization of angular momentum, deduce the expression for the magnetic moment of hydrogen atom in the ground state.

CBSE 2019

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CBSE 2019

The magnetic moment is

$m = IA$

But current is $I = \frac{e}{T} = \frac{ev}{2\pi r}$ ½ ½

Where $T = \frac{2\pi r}{v}$ *and the area,* $A = \pi r^2$ ½

$m = \frac{ev}{2\pi r}\pi r^2 = \frac{evr}{2}$ ½

But from Bohr's second postulate

$m_e vr = \frac{nh}{2\pi} = \frac{h}{2\pi}$ for n=1 ½

$vr = \frac{nh}{2\pi m_e}$

Hence the magnetic moment is

$m = \frac{e}{2}\frac{h}{2\pi m} = \frac{eh}{4\pi m}$ (Here n=1) ½

State Bohr's quantization condition of angular momentum. Calculate the shortest wavelength of the Bracket series and state to which part of the electromagnetic spectrum does it belong.

CBSE 2019

State Bohr's quantization condition of angular momentum. Calculate the shortest wavelength of the Bracket series and state to which part of the electromagnetic spectrum does it belong.

CBSE 2019

Electron revolves around the nucleus only in those orbits for which the angular momentum is some integral of h/2π. (where h is planck's constant) (Also give full credit it a student write mathematically mvr = $\frac{nh}{2\pi}$)	½
$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$	½
For Brackett Series, Shortest wavelength is for the transition of electrons from $n_i = \infty$ to $n_f = 4$ $\frac{1}{\lambda} = R\left(\frac{1}{4^2}\right) = \frac{R}{16}$	
$\lambda = \frac{16}{R}\,\mathrm{m}$	½
= 1458.5 nm on substitution of value of R [Note: Don't deduct any mark for this part, when a student does not substitute the value of R, to calculate the numerical value of λ]	
Infrared region	½

Calculate the orbital period of the electron in the first excited state of hydrogen atom.

CBSE 2019

Calculate the orbital period of the electron in the first excited state of hydrogen atom.

CBSE 2019

$$Radius\ \ r_n = \frac{h^2 \epsilon_0}{\pi m e^2} n^2$$	½
$$velocity\ \ v_n = \frac{2\pi e^2}{4\pi\varepsilon_0 h}\frac{1}{n}$$	½
$$Time\ period\ \ T_n = \frac{2\pi r_n}{v_n} = \frac{4\varepsilon_0^2 h^3 n^3}{m e^4}$$	½
For first excited state of hydrogen atom n=2	
$$T_2 = \frac{32\varepsilon_0^2\ h^3}{m e^4}$$	½
On calculation we get $T_2 \approx 1.22\ X\ 10^{-15}\ s.$ (However, do not deduct the last ½ mark if a student does not calculate the numerical value of T_2)	

Calculate the ratio of the frequencies of the radiation emitted due to transition of the electron in a hydrogen atom from its (i) second permitted energy level to the first level and (ii) highest permitted energy level to the second permitted level.

CBSE 2018

Define the distance of closest approach. An α-particle of kinetic energy 'K' is bombarded on a thin gold foil. The distance of the closest approach is 'r'. What will be the distance of closest approach for an α-particle of double the kinetic energy ?

CBSE 2017

Define the distance of closest approach. An α-particle of kinetic energy 'K' is bombarded on a thin gold foil. The distance of the closest approach is 'r'. What will be the distance of closest approach for an α-particle of double the kinetic energy ?

CBSE 2017

It is the distance of charged particle from the centre of the nucleus, at which the whole of the initial kinetic energy of the (far off) charged particle gets converted into the electric potential energy of the system. Distance of closest approach (r_c) is given by	1
$\frac{\quad}{\quad} \quad \frac{\quad}{\quad}$	½
'K' is doubled, becomes −	½

Write two important limitations of Rutherford nuclear model of the atom.

Write two important limitations of Rutherford nuclear model of the atom.

CBSE 2017

1. According to Rutherford model, electron orbiting around the nucleus, continuously radiates energy due to the acceleration; hence the atom will not remain stable.	1
2. As electron spirals inwards; its angular velocity and frequency change continuously; therefore it will emit a continuous spectrum.	1

Find out the wavelength of the electron orbiting in the ground state of hydrogen atom.

CBSE 2017

Find out the wavelength of the electron orbiting in the ground state of hydrogen atom.

CBSE 2017

Radius of ground state of hydrogen atom =0.53Å = 0.53 x	½
According to de Broglie relation	½
For ground state $n=1$	
$2 \times 3.14 \times 0.53 \times 10^{-10} = 1 \times$ m	½
	½

A 12·5 eV electron beam is used to excite a gaseous hydrogen atom at room temperature. Determine the wavelengths and the corresponding series of the lines emitted.

CBSE 2017

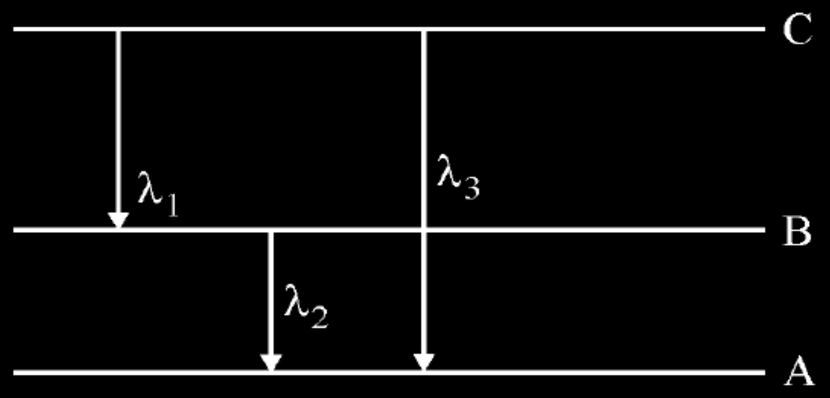
A 12·5 eV electron beam is used to excite a gaseous hydrogen atom at room temperature. Determine the wavelengths and the corresponding series of the lines emitted.

CBSE 2017

i.	We have $E_n \propto \frac{1}{n^2}$	½
ii.	∴ The energy levels are $-13.6\ \text{eV};\ -3.4\ \text{eV}; -1.5\ \text{eV}$ ∴ The 12.5 eV electron beam can excite the electron up to n=3 level only.	½
iii.	Energy values, of the emitted photons, of the three possible lines are $3 \rightarrow 1 : (-1.5 + 13.6)\,\text{eV} = 12.1\ \text{eV}$ $2 \rightarrow 1 : (-3.4 + 13.6)\,\text{eV} = 10.2\ \text{eV}$ $3 \rightarrow 2 : (-1.5 + 3.4)\,\text{eV} = 1.9\ \text{eV}$	
	The corresponding wavelengths are: 102 nm, 122 nm and 653 nm $\left(\lambda = \frac{hc}{E}\right)$	½ + ½

(i) State Bohr's quantization condition for defining stationary orbits. How does de Broglie hypothesis explain the stationary orbits ?

(ii) Find the relation between the three wavelengths λ_1, λ_2 and λ_3 from the energy level diagram shown below.



CBSE 2016

(i) Only those orbits are stable for which the angular momentum, of revolving electron, is an integral multiple of $\frac{h}{2\pi}$.

[Alternatively

$L = \frac{nh}{2\pi}$ i.e. angular momentum of orbiting electron is quantised.] ½

According to de Broglie hypothesis ½

Linear momentum $(p) = \frac{h}{\lambda}$

And for circular orbit $L = r_n p$ where 'r_n' is the radius of quantized orbits.

$$= \frac{rh}{\lambda}$$

Also $L = \frac{nh}{2\pi}$

$$\therefore \frac{rh}{\lambda} = \frac{nh}{2\pi}$$

$$\Longrightarrow 2\pi r_n = n\lambda$$ ½

∴ Circumference of permitted orbits are integral multiples of the wavelength λ

ii) $E_C - E_B = \frac{hc}{\lambda_1}$ .....(i)

$E_B - E_A = \frac{hc}{\lambda_2}$ .....(ii) ½

$E_C - E_A = \frac{hc}{\lambda_3}$ .....(iii)

Adding (i) & (ii)

$E_C - E_A = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}$.....(iv) ½

Using equation (iii) and (iv) ½

$\frac{hc}{\lambda_3} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2} \Rightarrow \frac{1}{\lambda_3} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

In the ground state of hydrogen atom, its Bohr radius is given as 5.3×10^{-11} m. The atom is excited such that the radius becomes 21.2×10^{-11} m. Find (i) the value of the principal quantum number and (ii) the total energy of the atom in this excited state.

CBSE 2016

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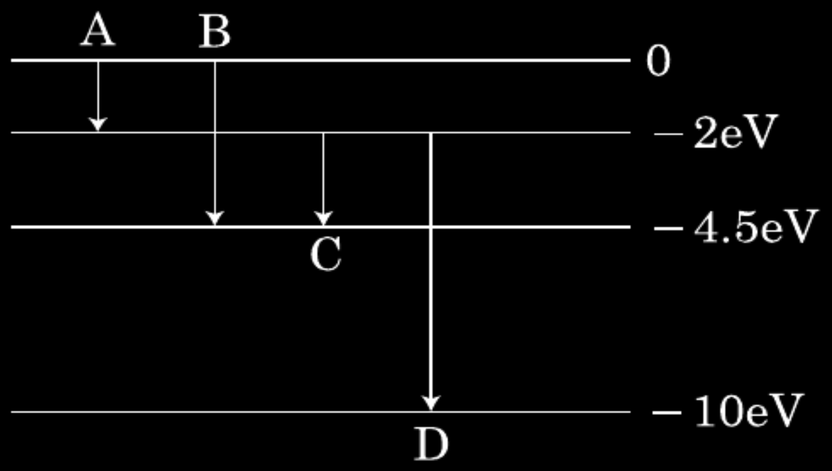
CBSE 2016

(i)	$r = r_0 n^2$	½
	$21.2 \times 10^{-11} = 5.3 \times 10^{-11}\ n^2$ implies $n = 2$	½
(ii)	$E = \frac{-13.6eV}{n^2}$	½
	$= \frac{-13.6eV}{2^2} = -3.4eV$	½
[Award ½ mark if the student just writes $E = E_1/4$]		

The energy levels of a hypothetical atom are shown below. Which of the shown transitions will result in the emission of photon of wavelength 275 nm ?

CBSE 2016



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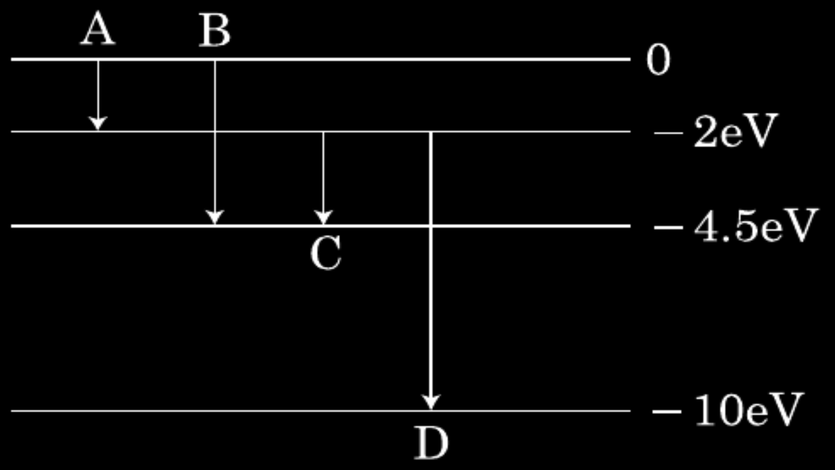
CBSE 2016



(i)	Energy of photon $= \frac{hc}{\lambda} = \frac{6.64 \times 10^{-34} \times 3 \times 10^{8}}{275 \times 10^{-9} \times 1.6 \times 10^{-18}}\ eV = 4.5\ eV$	½ + ½ + ½
(ii)	The corresponding transition is B	½

When is H_α line in the emission spectrum of hydrogen atom obtained ? Calculate the frequency of the photon emitted during this transition. CBSE 2016

When is H_α line in the emission spectrum of hydrogen atom obtained ? Calculate the frequency of the photon emitted during this transition. CBSE 2016

When the electron jumps from the orbit with n=3 to n=2 (Longest wavelength of the Balmer series / First line of the Balmer series)	1
$h\vartheta = E_3 - E_2 = \frac{E_1}{9} - \frac{E_1}{4}$ $= \frac{-5}{36}E_1 = \frac{-5}{36} \times (-13.6\ eV)$ $= \frac{5}{36} \times 13.6 \times 1.6 \times 10^{-19}\ J$	½
$\therefore \vartheta = \frac{5 \times 13.6 \times 1.6 \times 10^{-19}}{36 \times 6.63 \times 10^{-34}}$ Hz $\simeq 4.57 \times 10^{14}$ Hz.	½
(If the student just writes $\vartheta = \frac{-5}{36}\frac{E_1}{h}$, award ½ mark)	
(Alternatively, $\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = \frac{5}{36}R$	½
$\therefore \vartheta = \frac{c}{\lambda}$ $= c \times \frac{5}{36}R$ $= 3 \times 10^8 \times \frac{5}{36} \times 1.097 \times 10^7 Hz$ $\simeq 4.57 \times 10^{14} Hz$)	½

Calculate the wavelength of radiation emitted when electron in a hydrogen atom jumps from $n = \infty$ to $n = 1$.

CBSE 2016

Calculate the wavelength of radiation emitted when electron in a hydrogen atom jumps from $n = \infty$ to $n = 1$.

CBSE 2016

$\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{\infty^2}\right)$	½
$\therefore \lambda = \frac{1}{R}$	½
$= \frac{1}{1.097 \times 10^7}$ m	
$\simeq 9.116 \times 10^{-8}\ m$	
$\simeq 912\ A^0\ (91.2\ nm)$	1

Show mathematically how Bohr's postulate of quantization of orbital angular momentum in hydrogen atom is explained by de-Broglie's hypothesis.

CBSE 2016

Show mathematically how Bohr's postulate of quantization of orbital angular momentum in hydrogen atom is explained by de-Broglie's hypothesis.

CBSE 2016

de Broglie wavelength, $\lambda = \frac{h}{mv}$	½
For electron moving in the n^{th} orbit, $2\pi r = n\lambda$	½
$\therefore 2\pi r = \frac{nh}{mv}$	
$\therefore$ mvr $= \frac{nh}{2\pi}$ = L (orbital angular momentum)	½
This is Bohr's Postulate of quantization of orbital angular momentum.	½

A hydrogen atom initially in its ground state absorbs a photon and is in the excited state with energy 12.5 eV. Calculate the longest wavelength of the radiation emitted and identify the series to which it belongs. [Take Rydberg constant $R = 1.1 \times 10^7\ m^{-1}$]

CBSE 2016

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CBSE 2016

$$\frac{1}{\lambda_{max}} = R\ \left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$ ½

The energy of the incident photon = 12.5 eV

Energy of ground state = -13.6eV

∴, Energy after absorption of photon can be -1.1eV

This means that electron can go to the excited state $n_i = 3$. It emits photons of maximum wavelength on going to $n_f = 2$ i.e.

$$\frac{1}{\lambda_{max}} = \left(\frac{1}{2^2} - \frac{1}{3^2}\right) R$$ ½

$$\lambda_{max} = \frac{36}{5R}$$

$$= \frac{36}{5 \times 1.1 \times 10^7}$$

$$= 6.555 \times 10^{-7} \text{m} = 6555\ A^{\circ}$$ ½

It belongs to Balmer Series. ½

(2) Also award full 3 marks if the student writes that the energy of the excited state cannot be 12.5eV]

Use Bohr model of hydrogen atom to calculate the speed of the electron in the first excited state.

CBSE 2016

Use Bohr model of hydrogen atom to calculate the speed of the electron in the first excited state.

CBSE 2016

$v = \frac{nh}{2\pi mr}$ And $r = \frac{1}{k}\frac{n^2 h^2}{4\pi^2 me^2}$	½
So, $v = k\ \frac{2\pi e^2}{nh}$	
In first excited state n = 2	½
So velocity $v_2 = \frac{2\pi ke^2}{2h}$	½
$= 1.09 \times 10^6$ ms^{-1}	½
OR	
Velocity of electron, $v_n = \frac{1}{137}\frac{c}{n}$ In first excited state n =2 So velocity in first excited state (v_2)	1
$= \frac{1}{137}\frac{c}{2}$	½
$= 1.09 \times 10^6$ ms^{-1}	½

Calculate the de-Broglie wavelength of the electron orbitting in the $n = 2$ state of hydrogen atom.

CBSE 2016

Calculate the de-Broglie wavelength of the electron orbitting in the n = 2 state of hydrogen atom.

CBSE 2016

Kinetic Energy for the second state- $E_k = \frac{13.6eV}{n^2} = \frac{13.6eV}{4} = 3.4X1.6X10^{-19}J$	½
De Broglies wavelength $\lambda = \frac{h}{\sqrt{2mE_k}}$	½
$= \frac{6.63X10^{-34}}{\sqrt{2X9.1X10^{-31}X3.4X1.6X10^{-19}}}$	½
= 0.067nm	½

Define ionization energy.

How would the ionization energy change when electron in hydrogen atom is replaced by a particle of mass 200 times that of the electron but having the same charge ?

CBSE 2016

Define ionization energy.

How would the ionization energy change when electron in hydrogen atom is replaced by a particle of mass 200 times that of the electron but having the same charge ?

CBSE 2016

The minimum energy, required to free the electron from the ground state of the hydrogen atom, is known as Ionization Energy.	1
$E_o = \frac{me^4}{8 \in_o{}^2 h^2} i.e, .E_o \propto m$ Therefore, Ionization Energy will become 200 times	½

Calculate the shortest wavelength of the spectral lines emitted in Balmer series.

[Given Rydberg constant, $R = 10^7\ m^{-1}$]

CBSE 2016

Calculate the shortest wavelength of the spectral lines emitted in Balmer series.

[Given Rydberg constant, $R = 10^7\ m^{-1}$]

CBSE 2016

$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right)$	1
For shortest wavelength, n = α	
Therefore, $\frac{1}{\lambda} = \frac{R}{4} \Rightarrow \lambda = \frac{4}{R} = 4 \times 10^{-7} m$	½
	½